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Old but gold: Historical pathways and path dependence ☆

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ABSTRACT

Following the discovery of gold in 1694 in Brazil, pathways were constructed to connect coastal settlements to mining regions in the unpopulated interior. While these pathways initially facilitated the creation of road towns, their influence faded by the late nineteenth century. With the mid-twentieth-century demographic and industrial transition, regions with higher historical road density experienced renewed population growth and greater migrant inflows. We argue that this resurgence reflects the role of road towns in supporting early urbanization and structural transformation. Using an extended Rosen–Roback–Glaeser framework, we estimate strong agglomeration spillovers, consistent with Brazil's spatial economy exhibiting multiple steady states and possible historical path dependence.

1. Introduction

What drives the spatial distribution of economic activity and population? The literature extensively documents the role of location advantages and historical shocks in shaping economic geography. Some studies emphasize the persistence of location advantages as the main factor (*e.g.*, Davis and Weinstein, 2002), while others highlight the importance of historical events in determining where people settle (*e.g.*, Bleakley and Lin, 2012). A growing consensus suggests that "location fundamentals may be key when differences across locations are large, but other factors dominate as locations become more similar" (Hanlon and Heblich, 2022). Or, as put succinctly: "history matters only when it matters little" (Rauch, 1993).

In this paper, we study the long-run effects of historical roads on the distribution of population within Brazil. We contribute to the existing literature by revealing a dynamic pattern of spatial development that initially grows and then fades, only to resurge decades later, creating agglomeration economies consistent with path dependence driven by early urbanization and structural transformation. Our analysis centers on a pivotal event in the late seventeenth century: the sudden discovery of gold around 1694, which prompted the construction of "gold roads" connecting coastal settlements to the then unpopulated interior, which in turn facilitated the emergence of "road towns", a new type of urban settlement with a mostly non-rural mix of initial population. These routes eventually transformed into a national "mule road" network, vital for internal transportation until the introduction of railroads in the late nineteenth century.

While both types of roads eventually became obsolete with the advent of modern transportation in the twentieth century, our findings

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indicate that they remain associated with higher population density and economic activity today. Specifically, a 10-percentage-point increase in gold road density today is associated with an increase of approximately 25% in local population density, along with significant increases in nightlight intensity, urban population density, and urbanization rate. Mule roads also show a similar, though somewhat weaker, effect. These findings underscore the enduring impact of historical infrastructure on contemporary settlement patterns and economic activity.

Beyond modern outcomes, we find that the impact of the historical roads on population distribution initially faded by the early twentieth century, suggesting a return to the trend similar to the temporary effects observed in Davis and Weinstein (2002).[1] However, this influence reemerged in the mid-twentieth century, coinciding with the country's rapid urbanization, industrialization, and large-scale internal migration. Areas with higher gold road density saw faster population growth. This pattern arose not only because the remnants of these roads may have lowered migration costs but also because the original gold routes supported early urban development through road towns, which then acted as agglomeration seeds for early industrialization.[2] Mule roads also played a role in this later growth, although their influence was somewhat smaller.

To assess whether Brazil's spatial system admits multiple steady states, we estimate net agglomeration effects in an extended version of the Rosen–Roback–Glaeser framework derived by Allen and Donaldson (2020, 2022). Our estimates are consistent with an environment that admits multiple steady states, so that temporary historical shocks can alter which long-run equilibrium is selected, allowing for path dependence.[3] In our setting, this leaves open the possibility that the resurgence of historically advantageous regions could permanently reshape Brazil's long-term spatial configuration.

Gauging the effects of historical roads on economic development involves several challenges. First, acquiring relevant historical data is not a trivial task. We digitize gold roads from historical maps in Simonsen (1977), depicting routes connecting coastal settlements to gold mines discovered in Brazil's hinterlands in the last decade of the seventeenth century. In addition, for mule roads, we build a unique origin–destination network from nineteenth-century archival documents and government reports. Integrating these sources improves the external validity and provides robust empirical strategies. Using georeferenced trails, we compute road density as the area of a 5-kilometer buffer around the road divided by the region's total area. Outcome variables for population are constructed from censuses spanning 1920–2010.[4]

Second, inferring causality between historical roads and modern development requires careful attention to nonrandom factors, such as the influence of first-nature geography, pre-existing settlements, and strategic exploration on road placement. We combine two approaches. (1) We instrument historical roads with optimal least-cost paths (LCPs), computed from predetermined terrain-based travel costs, connecting locations associated with geologically predetermined mining sites to coastal settlements. (2) Following the inconsequential-units logic (Redding and Turner, 2015), we exclude pre-gold discovery settlements, including endpoints where gold was found, so that remaining locations were not themselves targets of strategic road placement.

Our identifying assumption is that, conditional on a rich set of geographic controls and flexible spatial trends, whether an "inconsequential" location lies on the cost-minimizing corridor is independent of unobserved determinants of modern outcomes, and affects outcomes only through realized historical roads. We provide four pieces of supporting evidence: (i) estimates are stable to sequential additions of the controls, thin-plate splines, and crop-suitability measures; (ii) results are similar when restricting comparisons to road municipalities and their immediate neighbors, or with other restrictions such as removing all towns where gold was discovered during the eighteenth century; (iii) placebo LCPs between historically unconnected settlements have no predictive power for modern development; and (iv) controlling for expected exposure in road networks, as suggested by Borusyak and Hull (2023), does not change the results.

This study engages with the broader debate on whether spatial distribution is shaped predominantly by location fundamentals or path dependence. While some research highlights the enduring role of location fundamentals (*e.g.*, Davis and Weinstein, 2002, 2008; Lee and Lin, 2017), others emphasize the significance of historical events in shaping current spatial configurations (*e.g.*, Redding et al., 2011; Bleakley and Lin, 2012, 2015; Maloney and Valencia Caicedo, 2016; Michaels and Rauch, 2017; Hanlon, 2017). Our findings speak to both views: the initial fading of road effects by the early twentieth century echoes the mean reversion documented in Davis and Weinstein (2002), while the mid-century resurgence is consistent with historical roads having long-lasting effects on population distribution through path dependence in Brazil—a country whose geography is less rugged than Japan's but more rugged and with higher transportation costs than the United States.

A rich strand of research documents how historical infrastructure and settlement patterns can exert lasting influence on modern economic geography. For instance, Jedwab and Moradi (2016), Jedwab et al. (2017) show that railroads and settler cities established during colonization in sub-Saharan Africa shaped the spatial distribution of population long after the original infrastructure waned. Similarly, Dalgaard et al. (2022) demonstrate that Roman roads strongly predict current road networks and economic outcomes in Europe today, a persistence that does not hold where wheeled transport was abandoned, while Paik and Shahi (2022) underscores how ancient nomadic migration routes in Asia persist in shaping current population density. On the settlement side, Bosker and Buringh (2017) and Cermeno and Enflo (2019) highlight the interplay between first-nature geography and second-nature agglomeration effects in Europe's urban evolution.

In the Brazilian context, Barsanetti (2021) finds that pre-Columbian trails continue to shape modern urbanization mediated by railroads but not paved roads, with early settlements playing a potential role, while Portugal and Barsanetti (2023) compare the divergent economic impacts of two of the first gold roads, highlighting the importance of historically driven trade intensities, and Barsanetti (2026) shows that being a railroad endpoint in the past predicts city size today. In addition, Reis (2014) documents that historical transportation costs gave rise to regional inequalities in the location of economic activity in Brazil, a finding consistent with the results in Marein (2022) for Puerto Rico.

These empirical findings resonate with broader theoretical insights from Allen and Donaldson (2022, 2020), who show that historical shocks can produce multiple spatial equilibria when agglomeration spillovers are sufficiently strong. Our paper advances this literature on several fronts by providing both a richer temporal analysis and a deeper

[1] Davis and Weinstein (2002) document that Japan's population distribution returned to its pre-World War II state after the destruction of physical capital during the war.

[2] We rule out the possibility that the placement of modern transportation infrastructure explains this dynamic. Notably, railroad infrastructure was already significant by 1920, when historical roads had no effect on population density. Additionally, less than 10% of the highway network had been constructed by 1960, yet the positive relationship between historical roads and population density was already apparent. Moreover, the coefficients associated with this relationship remain robust when controlling for the Minimum Spanning Tree network originating from Brasília, an exogenous predictor of highway expansion proposed by Morten and Oliveira (2024).

[3] At the same time, our estimates do not imply divergence away from a steady state once the economy is sufficiently close to one. Within a given basin of attraction, convergence is relatively fast: the estimated partial-equilibrium persistence parameter implies that, absent a change of equilibrium, less than 6% of an initial shock would remain after two centuries.

[4] The shapefiles for the historical pathways, together with the full replication package (data and code), are available at Baerlocher et al. (2026).

investigation of both the mechanisms underlying the contemporaneous outcome and the agglomeration spillover (see, *e.g.*, Lin and Rauch, 2022).

We also move beyond standard long-run outcome comparisons by tracking population changes across multiple census decades, unveiling a dynamic pattern wherein the influence of the historical roads dissipates by the early twentieth century, only to reemerge during Brazil's rapid mid-century urbanization and industrialization. This temporal lens allows us to pinpoint the roles of early urban formation, subsequent industrialization, and internal migration in generating the observed path dependence, highlighting that our findings capture not only current outcomes but also the channels that plausibly drive them. Finally, by embedding these reduced-form results in an extended Rosen–Roback–Glaeser framework, we quantify historical and contemporaneous agglomeration spillovers and show how these forces may generate multiple equilibria and reshape the spatial system in the long run.

In doing so, our study also contributes to the literature on estimating productivity spillover effects. First, we introduce a novel instrumental variable (IV) tailored to the Brazilian context, whereas much of the existing literature uses population density directly or instruments it using lagged population (*e.g.* Barufi et al., 2016; Chauvin et al., 2017; Ehrl and Monasterio, 2021; Dingel et al., 2021). A notable exception is Almeida et al. (2023), who apply a shift-share IV approach to estimate the agglomeration coefficient, though their focus is on different outcomes. Additionally, we are able to disentangle contemporaneous spillovers from historical spillovers, as in Allen and Donaldson (2020).

2. Gold and historical pathways

In this section, we briefly explore the history of urban agglomerations in Brazil, emphasizing the significance of the gold roads in facilitating early movements toward the interior and the development of the ground transportation network in the eighteenth and nineteenth centuries. We also introduce the main historical data sources and definitions used throughout the paper.

Following their initial contact in 1500, Portuguese colonizers established settlements along the Brazilian coast. For an extended period, they avoided venturing into the hinterlands due to the challenges posed by the indigenous population, geographical barriers, and high transportation costs.[5] In the late sixteenth century, Brazil had only three towns and fourteen villages, mostly scattered along the coastline where sugar was produced.[6] São Paulo, founded in 1554, was a notable exception, as it was located inland (Azevedo, 1956). The establishment of São Paulo required arduous expeditions to penetrate the country's interior, carving paths through dense rainforests and the coastal mountain range while navigating conflicts and negotiations with the indigenous population.[7]

The main incursions into the hinterlands consisted of Jesuit missions and slave-raiding expeditions to capture the indigenous population, known as *Bandeirantes*, who primarily followed indigenous trails and relied on enslaved individuals as porters (Abreu, 1998; Kok, 2009). Despite these efforts, the missions and *bandeirante* expeditions failed to significantly alter the population's concentration along the coast (Deffontaines, 1938; Morse, 1974). During the seventeenth century, 37 villages were established, with a few inland, but the majority still along the coast (Azevedo, 1956).

It was only when gold mines were discovered around 1694 in the state of Minas Gerais and later in Goiás and Mato Grosso, that the economy shifted from coastal sugarcane plantations to the mining regions of the interior.[8] Several pathways were established to connect the mines with coastal settlements, facilitating the transport of agricultural provisions to the mining areas and the shipment of precious metals back to the coast (Russell-Wood, 1984). During the gold mining era, the main transportation system was the *mule train*, which proved an excellent option to trail irregular pathways.[9]

To accurately depict the spatial distribution of settlements in colonial Brazil, we rely on two primary sources. First, Azevedo (1956) provides a comprehensive list of villages and towns, along with their foundation years, from 1500 to 1822. To complement this list and expand it to include other years, we use historical data on current municipalities made available by the Brazilian Institute of Geography and Statistics (IBGE). This information, presented as brief histories of municipalities and their administrative evolution, is collected through programmatic data extraction. This additional data enables us to identify settlements that were not officially classified as villages by the Portuguese crown.[10] Details about the data extraction procedure are available in Appendix A.

Fig. 1 illustrates the spatial distribution of settlements in Brazil, along with the gold roads, both before (1694) and after (1822) the gold cycle era. The gold roads' trace was georeferenced from Simonsen (1977). For context, we also highlight major rivers and the states where gold was discovered. In addition to towns and villages, the map includes districts, parishes, and other smaller settlements. In Panel A, we identify 48 villages and towns scattered along the Brazilian coast before 1694, as noted earlier, along with 15 other settlements also located on the coast. Notable exceptions include settlements in the north, along the Amazon River, and Jacobina in the interior of Bahia, which was founded in 1677 as part of expeditions searching for gold.

Fig. 1B shows the spatial distribution of agglomerations in 1822, the year of Brazil's independence, providing a snapshot of the landscape left by Portuguese colonization. This date also enables a meaningful comparison with 1694, as the administrative structure remained under the same political regime. By 1822, gold mines were largely depleted, and gold had lost much of its economic significance compared to a century earlier. Our data identifies 217 villages and towns, along with 200 smaller settlements. The eastern coast is densely populated with settlements, while those in the interior are concentrated along the gold roads, the Amazon River, and the far southern regions.

[5] In 1627, the Franciscan friar Vicente de Salvador famously likened the Portuguese presence on the coast to crabs scratching the shores (Salvador, 2010, p. 70). Later in the 1700s, the Benedictine friar Gaspar da Madre de Deus observed that the essential reason why the Portuguese had decided to settle only on the coast was transportation costs: King John III of Portugal (1502–1557) knew that all goods produced along the coast could be easily transported to Europe, while those of the hinterland would never get to the ports and if they did the costs would be such that the farmers would not sell for the prices offered to them (Deus, 1920, p. 179). Unlike the United States, Brazil has a 1500-kilometer mountain range that separates coastal areas from the hinterland (*Serra do Mar*), in addition to many other mountain ranges to traverse while traveling from the coast toward the interior (see, *e.g.*, Deffontaines, 1937; Momsen, Jr., 1963).

[6] We use the following terms to categorize settlements: towns (*cidades*), villages (*vilas*), and other settlements (*distritos*, *freguesias*, *povoados*, and *arraiais*). Towns and villages are similar to modern municipalities. Other settlements refer to communities lacking administrative autonomy.

[7] For instance, enslaved porters carrying 30-kilogram loads needed four to five days to complete the 70-kilometer round trip between the port city of Santos and São Paulo, a route that crossed the imposing *Serra do Mar* mountain range (Reis, 2023).

[8] The exact date of discovery is uncertain, but historical accounts suggest that gold was found between 1693 and 1695 in various parts of Minas Gerais (Boxer, 2003, p. 35). See Arroyo Abad and Palma (2021) for a comparative review of the discovery of gold in Brazil. The gold states are highlighted in the map below.

[9] See, for example, Schmidt (1959) and Goulart (1961) for the leading role of the mule in Brazil's historical development.

[10] For instance, relying solely on Azevedo (1956), the capital of Pernambuco, Recife, appears in our list only in 1709, when it was officially elevated to village status. However, historical records indicate that the settlement existed as early as 1561. Additionally, Azevedo (1956) omits several villages that the programmatic data extraction successfully identifies and includes.

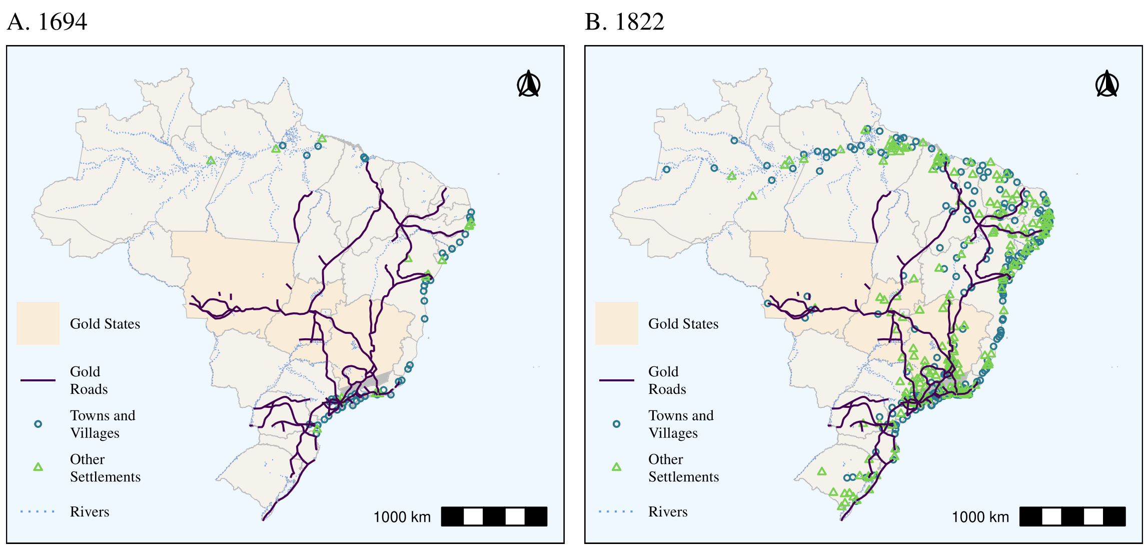

Fig. 1. Gold roads and population settlements.
Notes: Spatial distribution of settlements in Brazil before (1694) and after (1822) the gold cycle era. The gold states are Minas Gerais, Goiás, and Mato Grosso. Gold roads were georeferenced from maps in Simonsen (1977). Settlements and their classification are from Azevedo (1956), and the history of municipalities from IBGE.

The gold cycle in Brazil lasted for roughly fifty years, from 1700 to 1750 (Russell-Wood, 1984). From 1730 to 1875, mule trains dominated long-distance inland transport, and over time the gold roads evolved into a nationwide mule-road network, especially after the 1830s when pathways expanded rapidly (Morais, 2010). Throughout the main text, we focus on the effects of gold roads on the spatial distribution of population, while extensions and robustness checks, mostly reported in the appendix, examine the broader mule road network to assess the external validity of our results.

Our main variable of interest throughout the paper is gold road density, defined as the area within a 5-kilometer buffer around the gold roads within a region relative to the region's total area. Gold road density can be directly calculated for gold roads using the maps from Simonsen (1977) as depicted in Fig. 1. Additionally, we use the density of LCP traces as an IV for gold road density. The LCPs are calculated using detailed terrain-topology data to estimate the travel time required to traverse the landscape on foot, based on slope, following the methodology of Barjamovic et al. (2019). The underlying assumption is that, while animals such as horses and mules accompanied travelers, they did not impose significant constraints on travel time. Nor were they likely to increase speed, as historical accounts indicate that wheeled transportation was infeasible due to poor road quality and that many participants in the *mule trains* traveled on foot rather than riding because the mules were used mainly to carry goods.

In Fig. 2A, we display the LCPs for gold roads — together with their original traces — which we use to construct an IV for gold-road density. Fig. 2B shows the gold road density in Brazilian municipalities in 2010. Details regarding data sources and methodological procedures are provided in Appendix A.

This section detailed the development of early transport infrastructure in previously unpopulated areas, spurred by the discovery of gold, and its subsequent evolution into a broader network of ground transportation known as mule roads. These historical pathways remained the primary means of facilitating movement in the Brazilian hinterland until the late nineteenth century, when railroads were introduced. In the next section, we examine the relationship between these pathways and the current population distribution.

3. Historical pathways and the current population distribution

Having established the significance of historical roads, particularly the gold roads, in shaping the spatial distribution of settlements during the colonial era, we now examine whether these roads continue to influence population distribution today, despite having been rendered obsolete by the advent of modern transportation technologies.

Identification strategy. We model the relationship between historical roads and current agglomeration using the following linear regression equation:

$$y_i = \alpha_s + \beta \text{Road Density}_i + \mathbf{X}_i' \boldsymbol{\gamma} + \epsilon_i, \qquad (1)$$

where y_i represents a measure of agglomeration in municipality i in 2010. Specifically, y_i denotes the log-transformed values of population density, nightlight incidence, and urban population density, or the urbanization rate. Population density is defined as population over area, while urban population density is urban population over area. Urbanization rate is urban population over total population. Nightlight incidence is the median value of nightlights captured from satellites. The variable Road Density$_i$ captures the influence of historical roads measured as the area within a 5-kilometer buffer around the roads within a municipality relative to the municipality's total area. The term $\mathbf{X}_i$ indicates a set of geographical covariates, α_s denotes state fixed effects, and ϵ_i is the error term.

Our main concern is that, in a full-sample ordinary least squares (OLS) regression, the historical road density coefficient is biased by omitted variables related to first-nature advantages and to endogenous road placement and the related influence of settlements that existed before gold discovery. To address this, we combine an instrumental variables approach using optimal least-cost paths with a variant of the inconsequential units approach.

The optimal LCP IV strategy, similar to Barjamovic et al. (2019), leverages predetermined variation in topography to create an instrument independent of the existing economy, conditional on geography. The idea is that historical road builders should minimize costs based solely on topography, conditional on locational advantages that can be captured by observed geographic variables. Any actual deviations from these optimal paths are endogenously related to the existing economy

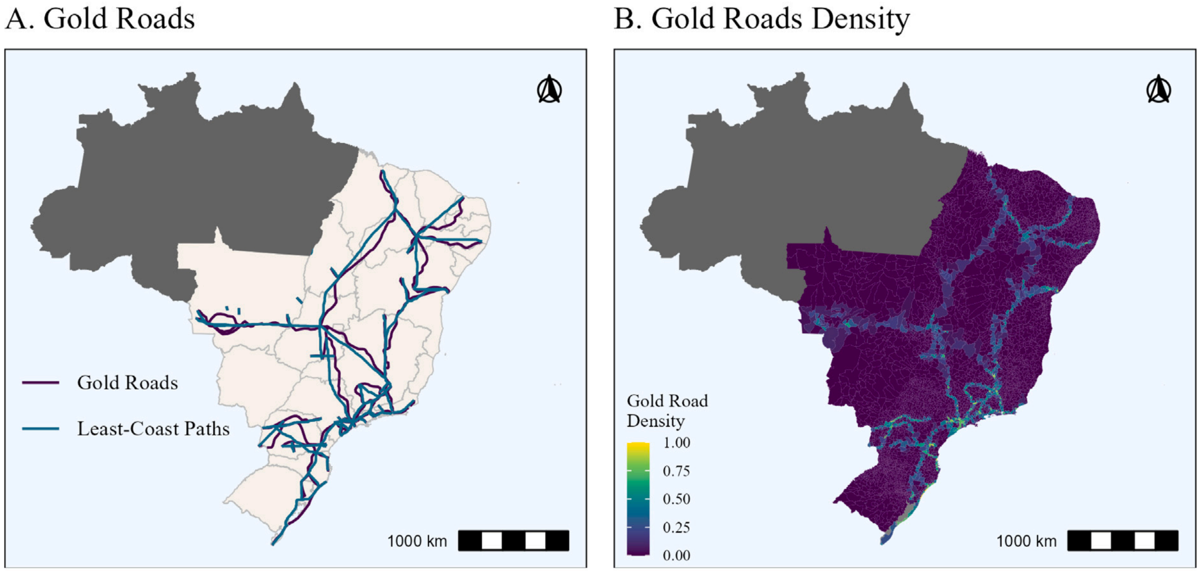

Fig. 2. Gold roads.
Notes: Panel A shows the traces of gold roads and their equivalent LCPs, while Panel B displays the gold road density defined as the area within a 5-kilometer buffer around the roads within a municipality relative to the municipality's total area. The LCPs minimize travel time by accounting for terrain topology, following the approach of Barjamovic et al. (2019). The gray-shaded areas, which are excluded from the analysis, represent Amazon rainforest states that mostly rely on river navigation: Amazonas, Acre, Rondônia, Roraima, Amapá, and Pará.

at the time the roads were created. The identifying assumption is that LCPs based on topography are as good as random, conditional on observed geographic characteristics. That is, among locations with similar geography, which particular locations happen to lie on the cost-minimizing route is determined by the geometry of connecting distant nodes rather than by local economic potential.

As expected, and shown in Fig. 1, proximity to rivers and the coast tends to attract settlements. Therefore, we control for the distance to rivers and the coast in our estimates. Additionally, we include median temperature, precipitation, terrain ruggedness, and elevation, as these factors affect local amenities and agricultural productivity. The area is also included as a control to account for the potential endogenous division of municipalities due to agglomeration effects. Finally, a second-order latitude–longitude polynomial is included to flexibly account for spatial trends in the location of municipalities. A placebo test that creates optimal LCPs between municipalities not connected by historical roads strengthens our confidence that, conditional on observed geography, the instrument is not capturing advantages arising from first-nature geography.

In addition, a distinctive feature of our setting is that the destinations of gold roads — the mining sites — were determined by prehistoric geology rather than by economic geography. Unlike infrastructure built to connect existing cities, where endpoint selection is endogenous, gold roads were built to reach mineral deposits whose locations were fixed by geological processes predating human settlement. In addition, Palma (2022) argues that the discovery of gold (i.e. timing and location of gold mines) in Brazil can be seen as an exogenous shock, that is, neither the Crown nor the overall population was expecting gold to be found where and when it was found. This strengthens the case for treating the network's overall structure as exogenous, not merely the specific routing between nodes.

We further restrict our sample to exclude endpoints where pre-established population settlements existed, as well as other pre-existing hinterland cities and villages built in the sixteenth and seventeenth centuries, including where gold was first found, that can drive endogenous strategic road placement. In our historical setting, this is less of a problem because, excluding the coast and the few interior settlements highlighted in Section 2, most of the country was not yet populated. Excluding the pre-existing settlements and endpoints, the remaining areas in the sample are inconsequential for route choice; they were chosen only because they lie on a convenient path (Redding and Turner, 2015).[11] Our regressions, therefore, compare inconsequential locations that were incidentally traversed by the historical roads against those that were not.

An additional concern we can address with our IV strategy is the presence of measurement error in our historical roads variable, given that we construct it using historical maps and documents. Historical documents can contain imprecision because of the technology of the time and errors by the authors. Furthermore, digitizing historical printed maps can also introduce distortions. In contrast, to build our IV, we use high-resolution data from satellites with a least-cost algorithm that yields paths with high accuracy, thereby mitigating potential bias toward zero in the road density coefficient.

The resulting two-stage least squares (2SLS) coefficient is thus the local average treatment effect (LATE) of increasing historical-road exposure for "inconsequential" municipalities that had their realized road density shifted by the terrain-predicted least-cost corridor.

Final dataset. Our final gold road sample excludes, in addition to pre-existing and endpoint municipalities, the municipalities in the northern states, which are mostly covered by the Amazon rainforest and where transportation is primarily by rivers (Amazonas, Acre, Rondônia, Roraima, Amapá, and Pará). The main sample consists of 5197 municipalities, with population densities ranging from 0.23 to 12,512 individuals per square kilometer and gold road density ranging from 0 to 1. This distribution is highly skewed, with more than 75% of the sample having zero gold road density. On average, municipalities exhibit a gold road density of 5%. Further details on the construction of the variables are provided in Appendix A and descriptive statistics are provided in Table A.1 in Section A.2. When restricting the sample to municipalities traversed by gold roads and their direct neighbors, the sample size reduces to 2088 observations. The average population density increases, indicating that we are sampling more developed

[11] Along these lines, Barsanetti (2026) emphasizes the enduring role of railroad endpoints in determining city size and "agglomeration shadows" in Brazil.

Table 1
Gold roads and current agglomerations.

	OLS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable: Population Density						
Gold Road Density	1.65***	1.01***	0.839***	3.30***	2.31***	2.22***
	(0.369)	(0.237)	(0.202)	(0.626)	(0.494)	(0.576)
Observations	5197	5197	2088	5197	5197	2,088
Dependent Variable: Nightlights						
Gold Road Density	1.17***	0.567***	0.507***	2.43***	1.47***	1.43***
	(0.222)	(0.132)	(0.119)	(0.408)	(0.293)	(0.380)
Observations	5197	5197	2088	5197	5197	2,088
Dependent Variable: Urban Pop. Density						
Gold Road Density	1.90***	1.18***	0.994***	3.78***	2.67***	2.54***
	(0.379)	(0.248)	(0.216)	(0.667)	(0.537)	(0.642)
Observations	5196	5196	2087	5196	5196	2,087
Dependent Variable: Urbanization Rate						
Gold Road Density	0.154***	0.101***	0.092***	0.293***	0.216***	0.185***
	(0.029)	(0.021)	(0.019)	(0.062)	(0.046)	(0.058)
Observations	5196	5196	2087	5196	5196	2,087
Cluster Groups:	404	404	260	404	404	260
Kleibergen–Paap F				133.88	130.74	94.796
Fixed Effects:	State	State	State	State	State	State
Geography Controls:		✓	✓		✓	✓
Only Neighbors:			✓			✓

Notes: Standard errors, clustered at the level of 1872 MCAs, are shown in parentheses. The sample includes municipalities as defined in 2010, excluding those in states heavily influenced by the Amazon River basin and municipalities with settlements that existed prior to 1694. *Population Density*, *Nightlights*, *Urban Population Density* are the log-transformed values of population over area, median nightlight incidence, and urban population over area, respectively. *Urbanization Rate* is the ratio of urban population to total population. *Gold Road Density* is defined as the area within a 5-kilometer buffer around gold roads within a municipality, relative to the municipality's total area. The 2SLS specification uses LCP density as an IV for gold road density. *Geography Controls* include the log-transformed median values of precipitation, temperature, elevation, terrain ruggedness, area, distance to rivers, and distance to the coast, along with a second-order latitude–longitude polynomial. *Only Neighbors* restricts the sample to municipalities intersected by gold roads and their first-order neighbors. *p<0.1,**p<0.05,*** p<0.01.

areas, and the distribution of gold road density becomes less skewed, with an average of 13%.

Main results. The main results are presented in Table 1. All estimates include state fixed effects, with standard errors clustered at the 1872 Minimum Comparable Areas (MCAs) level, where MCAs are aggregations of municipalities constructed by IBGE to ensure spatial consistency across census years. In Columns (4) to (6), the LCPs of gold roads is used as an instrumental variable in a 2SLS estimation. Except for Columns (1) and (4), the others control for geography, while Columns (3) and (6) restrict the sample to municipalities crossed by gold roads and their immediate neighbors. Overall, the results indicate a positive relationship between access to gold roads and population concentration, as measured by population density (Panel A), nightlights (Panel B), urban population density (Panel C), and urbanization rate (Panel D). Although the effect slightly diminishes with the inclusion of geographic controls, it remains both statistically and economically significant. This reduction likely reflects potential positive bias from location fundamentals. According to the specification in Column (6), a 10-percentage-point increase in road density is associated with a 24.9% increase in population density, a 15.4% rise in nightlight intensity, a 28.9% increase in urban population density, and a 1.85 percentage-point increase in urbanization rate.

As our model is exactly identified, the Kleibergen–Paap F statistic is equivalent to the effective first-stage F statistic proposed by Montiel-Olea and Pflueger (2013). In Table 1, we see that our F statistic substantially exceeds the critical value (37.42) corresponding to a 5% 2SLS bias (Montiel-Olea and Pflueger, 2013). Therefore, we can confidently apply the conventional 2SLS method with no concerns about weak instrument issues, as discussed by Andrews et al. (2019).

Moreover, the bias introduced by the OLS estimation is substantial, leading to an underestimation of the effect of historical roads. The larger 2SLS estimates relative to the OLS estimates may suggest the presence of measurement error in the historical map used to construct the gold roads, which would explain the downward bias in the OLS results. Since our instrument accounts for key topographical features, measurement error in the digitized gold roads likely causes naive OLS estimates to underestimate the true impact of the roads.

A natural concern in our setting is that historical gold roads may have generated spillovers or general equilibrium effects on nearby municipalities. If neighboring locations benefited indirectly from similar historical forces — through localized migration, trade linkages, or shared market access — our estimated coefficients could be attenuated, reflecting a partial-treatment contrast rather than the full effect of the historical road network. In principle, such spillovers could blur the distinction between treated and control areas and lead to conservative estimates of the long-run influence of gold roads.

Although spillovers and general equilibrium effects could, in principle, attenuate our estimates by partially treating nearby control units, the neighbors-only specifications in Table 1 Columns (3) and (6) show that this concern is limited. When the comparison is restricted to municipalities crossed by gold roads and their immediate neighbors, the coefficients remain large and precisely estimated. If spillovers or general equilibrium adjustments were substantial, this restricted sample would collapse the treated-control contrast and yield markedly smaller effects. The stability of the estimates therefore indicates that any such forces are too small to drive our results.[12]

[12] Additionally, when we turn to spatially aggregated units in Section 4 using MCAs, which group multiple municipalities and therefore absorb small-scale spatial interactions, the magnitude and persistence of the estimated effects remain stable, reinforcing that spillovers cannot account for our findings. In the historical panel specifications (Appendix Table C.1), the neighbors-only coefficients are even larger than those obtained in the full sample, indicating that proximity amplifies rather than attenuates the historical signal. Taken together, these patterns provide strong evidence that neither spillovers nor general equilibrium adjustments across municipal borders drive our results.

Robustness. In Appendix B, we demonstrate the robustness of our results through various extensions and alternative specifications. Specifically, we show that the results remain unchanged when we restrict the sample to exclude municipalities in which the seat is located within 100 kilometers of the coast. This case addresses concerns that the dynamics of coastal cities, which were the main population settlements before 1694, may drive our results. In this scenario, estimates are smaller but still statistically and economically significant. In addition, excluding also all municipalities that had gold discoveries in the eighteenth century leaves the results unchanged, indicating that our results are not about gold in itself. We also demonstrate that our results are robust to the use of the inverse hyperbolic sine transformation on our variable of interest. This transformation alleviates concerns related to the right-skewness of gold road density, which may include zero-valued observations.

Moreover, the effect remains positive and significant when our explanatory variable indicates whether road density is positive. Again, this mitigates concerns about the choice of measures for gold roads.[13] The results are similar but somewhat weaker within 1872 MCAs. We acknowledge there can be a potential presence of spatial correlation among the units, thus we cluster errors at the level of 1872 MCA. As an alternative approach, we also calculate spatially robust standard errors (Conley, 1999). Importantly, Conley standard errors are similar to the ones clustered at the level of 1872 MCAs, suggesting that clustered standard errors at this level already account for spatial correlation.

A remaining concern is that our estimates may reflect broad spatial patterns rather than local variation associated with historical transportation routes. Throughout the analysis, standard errors are clustered at the level of 1872 MCAs to allow for spatial correlation in the error term, and a robustness check using Conley standard errors yields virtually identical inference. Although all specifications already include a second-order polynomial in latitude and longitude to capture geographic gradients, we further assess robustness to more flexible forms of spatial dependence. Following Kelly et al. (2023), we estimate models that include a thin plate spline in latitude and longitude, allowing for smoothly varying spatial effects across the territory. The results are in Table B.2. At the municipality level — where spatial interactions and pendular movements across boundaries are likely stronger — the spline absorbs part of the variation, and the estimated coefficient on gold-road density decreases but remains large and statistically significant. In contrast, at the MCA level, where aggregation mitigates local spatial dependence, the coefficient is highly stable across all specifications. These results indicate that our findings are not driven by broad spatial trends or local spillovers in population density.

To assess whether the estimated relationship between historical gold road density and modern population density is driven by omitted variables or by the inclusion of specific controls, we examine how the coefficient changes as different sets of covariates are added sequentially in Table B.3. In both OLS and 2SLS specifications, the coefficient is remarkably stable. In 2SLS, the coefficient declines from 3.4 to 2.3 when area is added and remains around 2.2 once the full set of baseline controls is included. The Kleibergen–Paap F-statistics exceed 90 in all cases, indicating a strong first stage. Finally, adding crop-suitability variables (coffee, sugar, and wheat) as an additional robustness exercise leaves the results virtually unchanged. Overall, these results show that the positive long-run association between gold road density and modern population density is not driven by omitted geographic or climatic characteristics, and the magnitude of the effect is stable across a wide range of specifications.

Mule road extension, nonrandom exposure, and placebo test. We next extend our analysis beyond the gold roads to examine whether the long-run effects of historical transport infrastructure generalize to other networks. Specifically, we study the mule roads, which connected municipality seats across Brazil in the nineteenth century. This exercise broadens the geographical scope of our analysis and provides a valuable external validity test. The results, presented in Appendix G, confirm that areas historically traversed by mule roads also exhibit higher population density, nightlight intensity, and urban population density today. Although the estimated coefficients are smaller than those obtained for gold roads — consistent with the different historical roles of these networks — the positive association remains highly significant, suggesting that the enduring influence of historical infrastructure extends well beyond the context of the gold cycle.

The mule road analysis also helps address a potential concern that our earlier results may be confounded by geographic centrality. Because mule roads were constructed to integrate the national territory after independence, central locations may have been more likely to be traversed and may also enjoy higher market access today. To account for this potential source of nonrandom exposure, we follow Borusyak and Hull (2023) and include a measure of expected mule road density — constructed by simulating 1000 random realizations of the road network — as a proxy for centrality. The inclusion of this variable has little impact on the estimated coefficients, indicating that our results are not driven by central locations being more likely to host roads.

Finally, we implement a placebo test creating LCPs between historical settlements that were never actually connected. If our findings were driven solely by locational advantages correlated with optimal LCPs, these placebo connections would also be associated with higher modern population density. Instead, we find that the coefficients for placebo road density are small and become statistically insignificant once geography controls are included. This result underscores that the effects we document arise from the actual historical presence of roads, rather than from favorable geography alone. Taken together, these findings reinforce the external validity of our main results and provide strong evidence that historical infrastructure played a causal role in shaping Brazil's long-run spatial development.

4. Historical pathways and the dynamics of agglomeration

In this section, we examine the evolving impact of historical roads on population density over time. Specifically, we seek to determine whether these effects are strengthening or diminishing. A declining effect would suggest a historical shock operating within a highly persistent autoregressive spatial system, gradually fading over time. Conversely, a persistent or increasing effect would suggest the presence of multiple steady states, in which the historical shock may have initiated a transition toward a new equilibrium.

To maintain consistent boundaries over time, the units of observation in this analysis are the 1920 MCAs.[14] While this choice reduces the number of observations (there are 953 MCAs) and increases the geographical scale of analysis, it allows for a more consistent comparison across years. Reassuringly, the 2010 estimates based on the 1920 MCAs align closely with those in Table 1.[15]

For the census years from 1920 to 2010, our main variable of interest is population density. To extend the analysis further back in time, we utilize data on the existence of population settlements from 1570 to

[13] One point of concern may be that a large gold road density reflects more rugged terrain, leading to spurious interpretations. Although we control for terrain ruggedness, it is essential to show that the effects remain positive when using this alternative measure of gold road influence.

[14] In the Appendix Figure A.1, we plot a comparison of the units at the municipal and 1920 MCA levels.

[15] The 1920 MCAs are used because 1920 is the earliest year with reliable population data and a sufficient number of observations. Although earlier, the 1872 Census reduces the sample to 454 MCAs, significantly lowering statistical power. Results using 1872 MCAs are in Table C.2 in Appendix C. The 1890 and 1900 censuses are avoided due to known reliability issues (Oliveira and Simões, 2005).

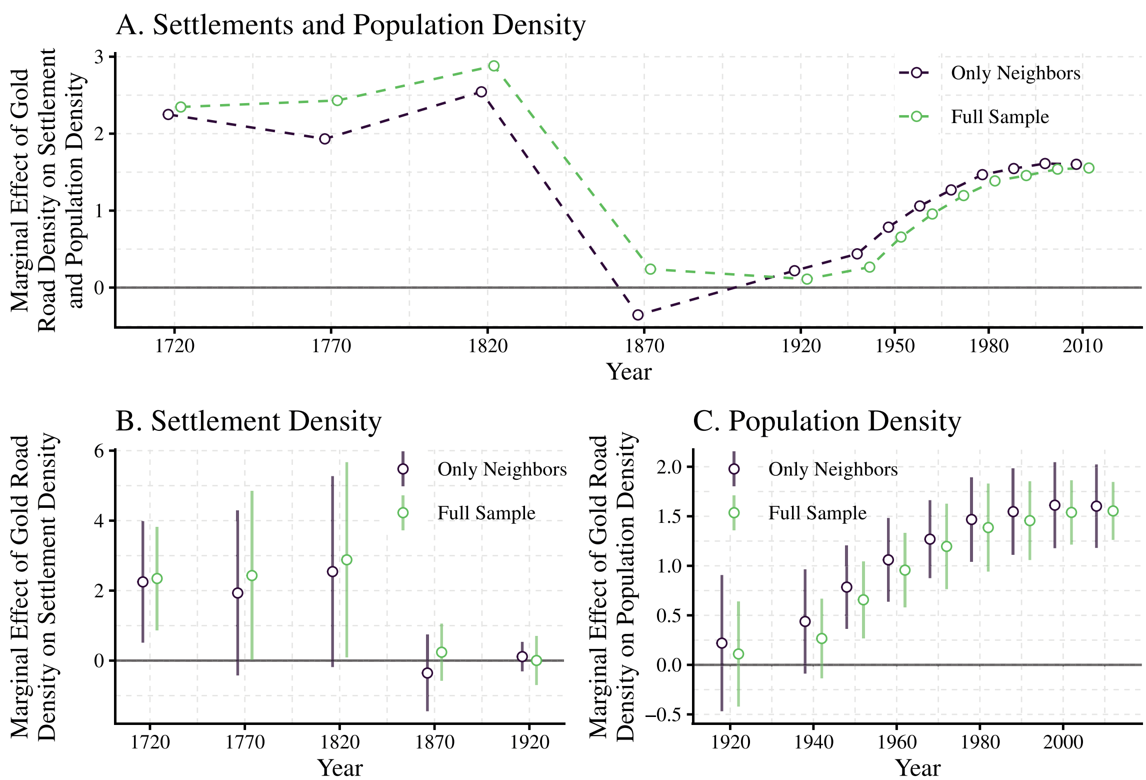

Fig. 3. Gold roads and settlement and population density over time.
Notes: The marginal effects of gold roads on settlement and population density are analyzed across different years using two samples and a 2SLS specification where LCP density is the IV for gold road density. The *Full Sample* consists of 1920 MCAs, excluding those in states heavily influenced by the Amazon River basin and MCAs with settlements predating 1694, resulting in 856 observations. The *Only Neighbors* sample includes only MCAs traversed by gold roads and their immediate neighbors, amounting to 604 observations. All estimates incorporate state fixed effects and geography controls. In all cases, the Kleibergen–Paap F is above 75. Point estimates are represented by circles, with 95% spatially corrected confidence intervals (Conley, 1999) shown as lines. Exact values are detailed in Table C.1.

1920, using the number of settlements as a proxy for agglomerations. Figure C.1 in Appendix C shows a strong positive correlation between population density and settlement density in 1920, which validates this analysis. For each year, at 50-year intervals, we calculate settlement density as the ratio of the number of settlements to the area for each 1920 MCA. Given the prevalence of zeros, particularly in earlier years, we apply an inverse hyperbolic sine transformation to this variable. In Panel E of Table C.3, we show that the results are qualitatively robust when we use a Poisson Pseudo-Maximum Likelihood regression to deal with zeros.

Fig. 3A shows the U-shaped relationship between gold road density and population density, recalling that before 1920 it is measured by settlement density. This distinction is emphasized in Panels B and C, where 95% spatially corrected confidence intervals are also added.[16] Coefficients are estimated using a 2SLS specification where LCP density is the instrumental variable for gold road density. The full sample includes 1920 MCAs, excluding the usual states heavily influenced by the Amazon River basin and MCAs with settlements predating 1694, resulting in 856 observations.[17] The “Only Neighbors” sample focuses on MCAs traversed by gold roads and their immediate neighbors, totaling 604 observations. All estimates account for state fixed effects and geography.

Gold roads exhibit positive effects on settlement density for over a century and a half. The peak of the gold cycle occurred during the first half of the eighteenth century, and the prominent role of gold roads persisted until the end of the colonial period in 1822. These findings align with the spatial distribution of settlements shown in Fig. 1B. Following independence, the focus on national integration spurred an increase in the number of settlements, independent of gold roads, such that by 1870, these roads were no longer associated with a higher density of settlements.[18] This trend continued into 1920.[19]

Data on population density agrees with the lack of effect of gold roads on population distribution in the first years of the twentieth century. In 1920, the effect is minimal, with a coefficient of less than 0.3 and statistically insignificant. However, the influence of gold roads on population density becomes increasingly pronounced over the course of the century. By 1940, the point estimate rises but remains below 0.5 and does not achieve statistical significance, even at the 10% level. The effect becomes statistically significant for the first time in 1950, with the coefficient peaking at approximately 1.5 in 1990,

[16] Given the limited number of observations per cluster when clustering errors at the level of 1872 MCAs, we present spatially corrected confidence intervals following Conley (1999), using a cutoff of 190 kilometers.

[17] A full set of descriptive statistics is available in Panel C of Table A.1.

[18] According to our data, the number of settlements rose from 102 in 1720 to 407 in 1820, and then to 1055 in 1870.

[19] In Table C.3 we also report estimates for the years prior to the gold discoveries: 1570, 1620, and 1670. The results generally suggest that the influence of gold roads gradually increased over this period. However, these coefficients are not directly comparable to those for later years because we cannot exclude pre-gold-era settlements. As a result, part of the estimated effect may simply reflect that gold roads were connected to established settlements along the coast.

followed by only a slight increase in subsequent decades. Between 1940 and 2010, the effect of gold roads on population density grows by a factor of three. Estimation details are provided in Table C.1. Similar estimates for mule roads are also presented in Table G.3. While the pattern of increasing influence is evident for mule roads — particularly after 1960 — the coefficients are statistically significant only starting in 1970. This aligns with earlier findings, which highlight the secondary importance of mule roads compared to gold roads.

The recent revival of the importance of gold roads is somewhat unexpected. The first half of the twentieth century was dominated by railroad transportation, although mule roads still played a role in connecting cities. As noted in Simonsen (1977), mules continued to be vital for intercity connections in Brazil even as late as 1937. However, their freight rates were significantly higher — ranging from three to eight times those of railroads — during the period between 1860 and 1915, as documented by Reis (2023). The second half of the century, however, saw the rapid expansion of highway networks, which diminished the importance of mule trains, a point we will explore in greater detail below. Given these historical developments, one might have expected the influence of historical roads to decline over the century.

Overall, the trend depicted in Fig. 3 reflects a spatial system where the effects of historical shocks gradually dissipate until 1870. Specifically, early agglomerations formed along the gold roads did not generate additional advantages, such as more capital or stronger institutions, that could spur further population growth. By 1870, settlement density in these areas had not become significantly higher or lower than in other regions. This trend continued into the first half of the next century. However, the effect of gold roads on population density reverses in the latter half of the twentieth century, showing a positive and growing impact.

The limited impact of gold roads on sustained population growth can partly be attributed to the transitory nature of the gold cycle and the poor quality of the roads themselves. Population movements to the hinterlands were not driven by reduced transportation costs induced by these roads but rather by the high value-to-weight ratio of gold and precious minerals. This characteristic rendered transportation costs negligible for commodity outflows but not for the inflow of goods needed by local populations. For instance, prohibitive transportation costs often triggered speculative crises in the supply of foodstuffs: The prices of foodstuffs in Ouro Preto, one of the main mining centers, were 10 to 30 times higher than in São Paulo at the beginning of the eighteenth century (Reis, 2023). With the depletion of gold toward the end of the eighteenth century, the issue of high transportation costs became more acute. Consequently, Brazil evolved into what is often described as an archipelago of self-sufficient islands lacking integration (Deutsch, 1996).

A limitation of this analysis is the absence of data on population, although it is plausible that, prior to industrialization, the number of settlements serves as a reasonable proxy for population. To address this issue, we assign different weights to various types of settlements. In Panels C and D of Table C.3, we demonstrate that the results remain consistent when assigning towns and villages weights ten times larger than those of other settlements.[20]

In the following sections, we delve deeper into the puzzling "revival" of the advantages associated with gold roads in the mid-twentieth century. Notably, we rule out the possibility of direct benefits from these roads, as any such advantages diminished in the nineteenth century, and there is no plausible reason to believe that individuals simply started reusing these roads a century later without any associated major change in the environment. Instead, we propose that the structural changes associated with the emergence of a modern economy before 1950 triggered the resurgence of latent advantages tied to the gold roads and their settlements. Additionally, we investigate whether this new economic shock led to strong or weak agglomeration spillovers, shedding light on the equilibrium properties of this spatial system.

5. Connections with modern transportation

One possible interpretation of the patterns observed earlier is the connection between gold roads and the development of modern transportation networks. In other words, while gold roads may no longer provide direct benefits to nearby areas in recent times, they might have influenced the spatial distribution of modern transportation infrastructure. In this section, we explore this hypothesis by analyzing the timing of the observed effects alongside the emergence and expansion of modern transportation technologies, as illustrated in Fig. 4. Subsequently, we assess whether the observed results persist after accounting for the presence of modern transportation infrastructure.

In Fig. 4A, the evolution of railroad length aligns with historical accounts indicating that, during the latter half of the nineteenth century and the early twentieth century, the Brazilian central government actively promoted a railroad network to foster national integration. By 1920, over 60% of the total railroad network was already constructed, with this figure exceeding 90% by 1946. Additionally, Panel B shows that the significance of railroads for transporting goods was substantially higher in the early decades of the twentieth century than in later periods. Consequently, if gold roads influenced population density through the placement of railroads, we would expect to observe strong effects early in the century that diminish over time. However, the observed pattern contradicts this expectation, as the effects intensify later in the century.

The evolution of highways aligns more closely with our observations. In 1940, only a limited number of highway kilometers connected key cities, such as São Paulo and Rio de Janeiro, accounting for approximately 0.5% of the total highway network today. By 1950, this share had risen modestly to 1.4%. In 1951, the central government began prioritizing motorized vehicle roads as the primary strategy for national integration. However, substantial expansion occurred only during the second half of the 1960s and throughout the 1970s. According to our data, by 1960, only about 10% of the highway system had been constructed, but this share increased to 60% by 1980. Furthermore, Panel B highlights a significant inflection point in the late 1970s, marked by a sharp increase in the number of automobiles per capita.

Nonetheless, the expansion of paved roads and highways shows minimal overlap with the historical gold and mule roads, except in areas where these routes clearly represented the optimal path for any mode of transportation. Overall, these historical pathways were not well-suited for modern transportation needs. Even the "Royal Road", the most famous historical route connecting the mining regions of Minas Gerais to the court in Rio de Janeiro, demonstrates limited integration into modern infrastructure. Currently, only 20% of the Royal Road is paved, while 74% remains dirt roads, and 6% consists of trails.[21]

To further illustrate the limited overlap between historical and modern networks, we provide in the Appendix Figure D.1 a series of maps that overlay the gold-road system with the evolution of Brazil's modern

[20] However, the size of towns and villages likely varied significantly within this category. Azevedo (1956) notes that *Rio de Janeiro* and *Salvador* had populations of around 100,000 by the end of the colonial period. Medium-sized settlements, with populations ranging from 10,000 to 30,000, included cities like *Cuiabá*, *São Paulo*, and *Recife*. Smaller centers, with populations between 5000 and 10,000, were common in the gold-producing regions, such as *Mariana* and *Diamantina*.

[21] We consider a total of 1780 kilometers of the *Novo*, *Velho*, *Diamantes*, and *Sabarabuçu* paths. See https://institutoestradareal.com.br/en/ for detailed maps and photographs.

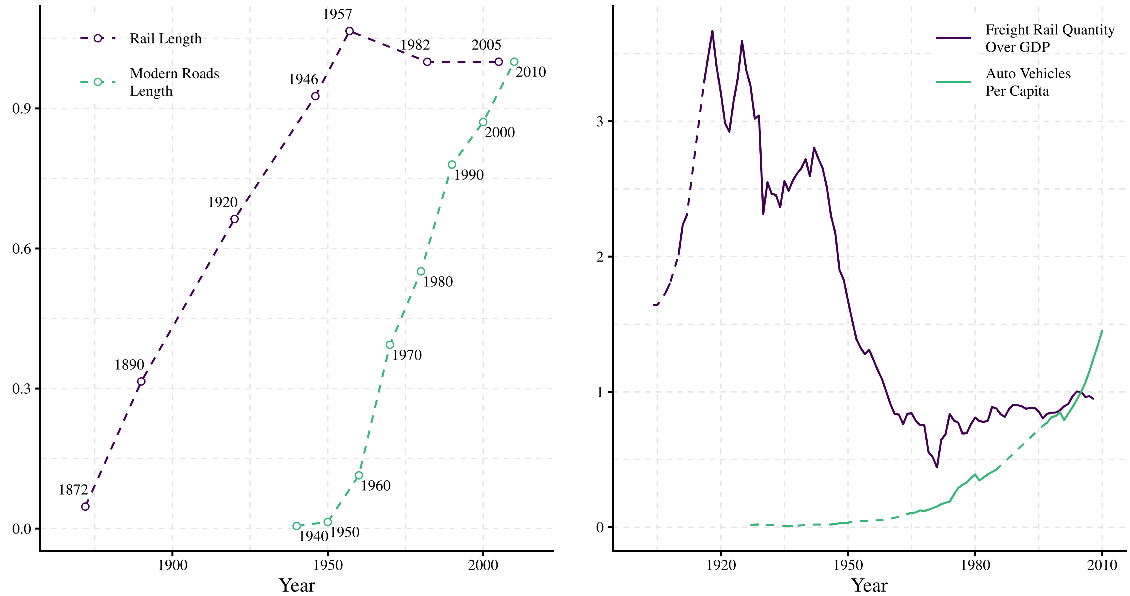

Fig. 4. Evolution of modern transportation.
Notes: Panel A illustrates the length of railroads and modern highways as a proportion of their respective lengths in the final year of the sample. Panel B displays the ratio of the quantity of goods (in tons) transported by railroad to real GDP, as well as the number of automotive vehicles per capita, both normalized to their 2005 values. Dashed lines represent linear interpolations between data points.

highway network from 1960 to 2010. The maps in Figure D.1 show that the major twentieth-century highways follow routes optimized for national integration rather than retracing the historical alignments, and only isolated stretches coincide with the gold cycle paths. This visual evidence reinforces the conclusion that modern infrastructure does not mechanically replicate the historical network, reducing concerns that our estimates are driven by contemporary road placement rather than long-run persistence. Although informative, this descriptive comparison has limitations such that we complement the maps with an estimation strategy that controls for modern infrastructure in a way that avoids bad-control problems, as described below.

To evaluate whether the effect of gold roads on population density is influenced by the presence of modern highways, we would include modern highway density as a control variable in our regressions. A key concern with this approach is that the placement of modern highways may be indirectly influenced by historical pathways through their impact on population density, potentially making highway density a "bad control". To mitigate this concern, we instead control for the distance to the Minimum Spanning Tree (MST) network emanating from Brasília—Brazil's capital since 1960 (Bird and Straub, 2020; Morten and Oliveira, 2024). As noted by Morten and Oliveira (2024), the Brazilian highway network was designed with the primary goal of national integration, with Brasília serving as its central hub. The radial highways originating from Brasília and connecting to key state capitals played a critical role in reducing travel times across the country.

To address the issue of endogenous highway placement, Morten and Oliveira (2024) divided Brazil into eight pie-shaped regions and constructed a predicted road network using LCPs that connect Brasília to the state capitals within these regions. This predicted network is referred to as the MST network. By using travel time along the MST network as an instrument for travel time along the actual highway network, Morten and Oliveira (2024) provide a framework to isolate the exogenous components of highway placement. Building on this MST-based approach, our analysis incorporates controls derived from this network to address potential biases and produce more robust estimates of the effect of gold roads on population density.

The results are presented in Table 2. These regressions replicate Columns (2) through (9) of Panels A and B in Table C.1, with the addition of the log-transformed distance to the MST network as a control.[22] As expected, MCAs located farther from the MST network exhibit lower population densities, with this negative correlation becoming stronger over time. Crucially, including this control has minimal impact on the coefficients associated with the gold roads. If anything, the magnitude of these coefficients slightly increases, reinforcing the robustness of the observed effects of gold roads on population density shown in Fig. 3, even after accounting for the MST network.

For comparison, Table D.1 presents the results when we control for highway density instead of distance to the MST network. Initially, highway density shows no significant correlation with population density in 1940 and 1950, likely reflecting the limited extent of the highway network at the time. However, starting in 1960, this correlation becomes positive and progressively stronger over the subsequent decades. Notably, while the importance of highway density increases, the effect of gold roads on population density diminishes but remains both statistically and economically significant. For example, in 2010 — the most extreme case — the coefficient for gold roads decreases from 1.55 to 1.07. The effect on the mule road coefficient moves in the opposite direction, increasing slightly when highway density is included as a control.

In Appendix G, we present additional results using mule roads. Table G.4 shows that the correlation between MST distance and population density is somewhat weaker and statistically insignificant in the mule roads sample, providing further evidence of the reduced statistical power of this smaller sample. Nevertheless, the reduction in the coefficient linking mule roads to population density over time is larger, though still positive and significant, compared to the gold roads case. This suggests that the overlap between mule roads and modern transportation infrastructure may not be negligible.

[22] We use the log-transformed distance to the MST network following Barsanetti (2024), who leverage the MST network to account for highway expansion. When we instead use the MST network density, the results remain consistent.

Table 2
Gold roads, modern roads and agglomerations over time.

	Years							
	1940	1950	1960	1970	1980	1990	2000	2010
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A - Full Sample - Dependent Variable: Population Density								
Gold Road Density	0.290	0.681***	0.982***	1.23***	1.43***	1.51***	1.59***	1.61***
	(0.248)	(0.253)	(0.271)	(0.291)	(0.316)	(0.337)	(0.339)	(0.346)
Dist. to MST	−0.066**	−0.067**	−0.072**	−0.088**	−0.117***	−0.141***	−0.155***	−0.162***
	(0.032)	(0.031)	(0.034)	(0.035)	(0.038)	(0.040)	(0.042)	(0.043)
Observations	856	856	856	856	856	856	856	856
Kleibergen–Paap F	81.431	81.431	81.431	81.431	81.431	81.431	81.431	81.431
Panel B - Only Neighbors - Dependent Variable: Population Density								
Gold Road Density	0.478*	0.825***	1.10***	1.32***	1.53***	1.62***	1.69***	1.69***
	(0.276)	(0.268)	(0.276)	(0.279)	(0.314)	(0.355)	(0.369)	(0.386)
Dist. to MST	−0.088**	−0.089***	−0.094**	−0.109***	−0.141***	−0.164***	−0.180***	−0.186***
	(0.034)	(0.034)	(0.039)	(0.041)	(0.045)	(0.048)	(0.052)	(0.055)
Observations	604	604	604	604	604	604	604	604
Kleibergen–Paap F	68.067	68.067	68.067	68.067	68.067	68.067	68.067	68.067

Notes: Spatially corrected standard errors are shown in parentheses. All columns report 2SLS estimates, using LCP density as an IV for gold road density. Each specification includes state fixed effects and geography controls, including the log-transformed median values of precipitation, temperature, elevation, terrain ruggedness, area, distance to rivers, and distance to the coast, as well as a second-order latitude–longitude polynomial. The *Full Sample* consists of 1920 MCAs, excluding those in states predominantly influenced by the Amazon River basin and MCAs with settlements predating 1694. The *Only Neighbors* sample includes only MCAs traversed by historical roads and their immediate neighbors. *Population Density* is calculated as the log-transformed population per unit area. *Gold Road Density* is defined as the proportion of an MCA's area within a 5-kilometer buffer surrounding gold roads. *Dist. to MST* is the log-transformed distance from the MCA's centroid to the Minimum Spanning Tree from Brasília computed by Morten and Oliveira (2024). *p<0.1,**p<0.05,*** p<0.01.

The results presented in this section challenge the hypothesis that modern transportation infrastructure is the primary mechanism linking gold roads to population density in recent times. In the next section, we investigate an alternative explanation proposed by historians, which suggests that towns established along historical pathways may have conferred enduring advantages that facilitated modern development.

6. Road towns and the seeds of agglomeration

Historians suggest that the primary benefit of the ground transportation system created from the gold economy lies not in the roads themselves but in the type of settlements established along these pathways. These distinctive settlements were more numerous and widespread than before, representing a new urban configuration. While estate-based settlements were dispersed and lacked a discernible urban center, the so-called "road towns" featured a main street lined with shops, cattle ranches, fairs, inns, and hotels. Naturally, these road towns attracted a different type of settler compared to agriculture-based towns, including craftsmen, workmen, merchants, and innkeepers, resulting in a distinctly different initial population mix (Deffontaines, 1938; Morse, 1974).[23]

There is no shortage of examples connecting the roads to modern settlements. Campinas, founded by an expedition connecting São Paulo to the mines in Goiás in 1722 (Rossetto, 2006), boasted a density of manufacturing workers four times larger than the national average in 1920. Piracicaba has a similar history: Although the area was first explored because of the Piracicaba River, with the first accounts dating back to 1693, it was only with the construction of the road leading from São Paulo to the mines in Mato Grosso that the area received its first settlers (IBGE, 1957). By 1920, Piracicaba had a manufacturing worker density two times the national average. In Minas Gerais, the founding of Santos Dumont and Juiz de Fora were both direct results of the creation of a new Royal Road path (*Caminho Novo*) to connect the Court in Rio de Janeiro to the mines, functioning as commercial and resting posts (IBGE, 1957). Unlike the São Paulo case, these locations are not tied to the coffee boom. By 1920, Juiz de Fora had a density of manufacturing workers three times the national average, while Santos Dumont had an average density but five times the national median.

[23] Deffontaines (1938) also argues that road towns were more stable than the mining towns themselves, as "mining colonization left only a devastated country strewn with dead or lethargic towns".

Table 3 examines whether gold roads shaped the early sectoral composition of local economies. Panel A shows that MCAs with higher gold-road density had a lower share of agricultural workers and a higher share of services and manufacturing workers in 1920, consistent with the road-town hypothesis. In Panel B, we formally assess this channel through a mediation analysis. We estimate 2SLS regressions in which gold-road density affects 2010 population density both directly and indirectly through its effect on the 1920 industrial composition. For each sector, we compute the Sobel test (Sobel, 1982), a standard approach to evaluate mediation effects, which assesses whether the product of the "first-stage" coefficient (from gold roads to the mediator) and the "second-stage" coefficient (from the mediator to the outcome) is statistically different from zero. A significant Sobel statistic indicates that the mediator transmits part of the total effect. The results show that 30%–45% of the long-run impact of gold-road density on modern population density is mediated through 1920 employment shares, particularly through the decline in agricultural specialization and the rise of services. These findings imply that gold roads shaped Brazil's spatial development through a road town mechanism that fostered early industrial and service-based activity, which then became self-reinforcing over time.

The hypothesis is that road towns did not hold much significance in an agrarian economy characterized by a small and sparse population. Nonetheless, as the population grew and the focus of production shifted toward non-agricultural activities, these settlements became attractive spots for migrants. Fig. 5 illustrates the substantial changes in the Brazilian socioeconomic structure over the twentieth century. In 1920, the population was only 16% of what it was in 2010, with rapid growth occurring during the 1950s and 1960s when population growth peaked at around 3% per year.

Internal migration accelerated in the 1930s, making the early dislocations caused by the sugar and gold economies appear less consequential (Furtado, 1968). This shift was precipitated by the structural transformation that began during the Great Depression years (Wagner and Ward, 1980). Over the century, population density increased substantially, rising from an average of 24 individuals per square kilometer in 1920 to 130 individuals per square kilometer in 2010, using 1920 MCAs as observations. This trend is reflected in the rapid increase in urbanization rates from the 1950s to the 1980s. In the 1940s, the urbanization rate grew by only 3 percentage points, whereas in the subsequent decades it increased by approximately 10 percentage points per decade.

Table 3
Gold roads and industrial composition in 1920.

	Full sample			Only neighbors		
	Agriculture (1)	Manufacturing (2)	Services (3)	Agriculture (4)	Manufacturing (5)	Services (6)
Panel A - Dependent Variable in Column						
Gold Road Density	−0.221***	0.110***	0.112***	−0.220***	0.097***	0.122***
	(0.057)	(0.029)	(0.034)	(0.027)	(0.020)	(0.018)
Observations	856	856	856	604	604	604
Kleibergen–Paap F	81.564	81.564	81.564	75.872	75.872	75.872
Panel B - Dependent Variable: Population Density 2010						
Gold Road Density	1.08***	1.36***	0.829**	1.30***	1.53***	0.910**
	(0.368)	(0.379)	(0.335)	(0.436)	(0.429)	(0.400)
Mediator (Column)	−2.15***	1.75**	6.48***	−1.37**	0.721	5.65***
	(0.562)	(0.880)	(0.721)	(0.674)	(0.952)	(0.926)
Observations	856	856	856	604	604	604
Kleibergen–Paap F	80.651	80.825	80.881	67.000	67.890	63.948
Sobel Test t	2.72	1.76	3.09	1.98	0.749	4.56
Share Mediated	0.306	0.123	0.467	0.188	0.044	0.432

Notes: Spatially corrected standard errors are shown in parentheses. All columns report 2SLS estimates, using LCP density as an IV for gold road density. Each specification includes state fixed effects and geography controls, including the log-transformed median values of precipitation, temperature, elevation, terrain ruggedness, area, distance to rivers, and distance to the coast, as well as a second-order latitude–longitude polynomial. The *Full Sample* consists of 1920 MCAs, excluding those in states predominantly influenced by the Amazon River basin and MCAs with settlements predating 1694. The *Only Neighbors* sample includes only MCAs traversed by historical roads and their immediate neighbors. The dependent variable is indicated in the column and measures the share of the population working in a given industry divided by the population working in all industries. *Gold Road Density* is defined as the proportion of a municipality's area within a 5-kilometer buffer surrounding gold roads. *p<0.1,**p<0.05,*** p<0.01.

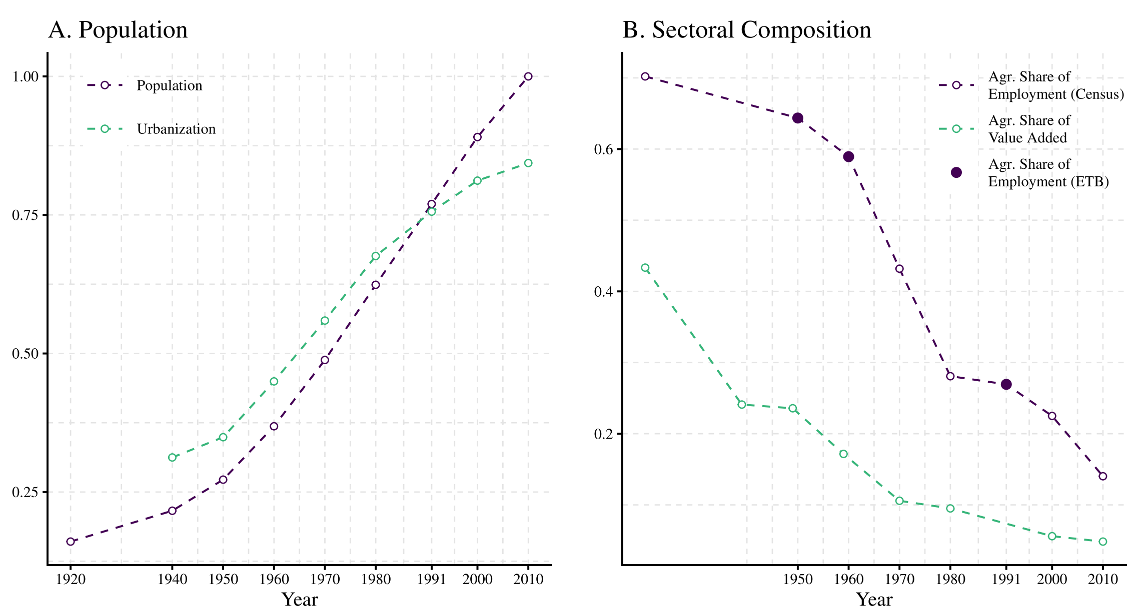

Fig. 5. The structural transformation of the Brazilian economy.
Notes: Panel A illustrates the population size relative to 2010 and the share of population living in urban areas. Panel B displays the share of value-added and employment allocated in the agricultural sector. There are two sources of employment data: the Brazilian Census and the Economic Transformation Database. Dashed lines represent linear interpolations between data points.

Fig. 5B illustrates the rapid process of industrialization, particularly between 1960 and 1980, when the share of workers in the agricultural sector decreased from about 60% to less than 30%.[24] The evolution of the share of value added from agriculture reveals a similar trend, with a significant decline first between 1920 and 1940, and then again from 1950 to 1970.

The overall pattern highlights the transition from an economy based on raw material exports to one that, by 1930, adopted an import substitution paradigm. The large influx of rural–urban migrants resulting from this shift found road towns to be attractive destinations, likely due to their already established urban infrastructure and well-developed service sector. Fig. 6 supports this hypothesis. Panel A illustrates the effect of gold road density on population growth across different decades, while Panel B focuses on migrant density. The coefficients are derived

[24] Whenever possible, the data presented are from the decennial census. We supplement the information for 1950, 1960, and 1991 using the Economic Transformation Database (Kruse et al., 2023).

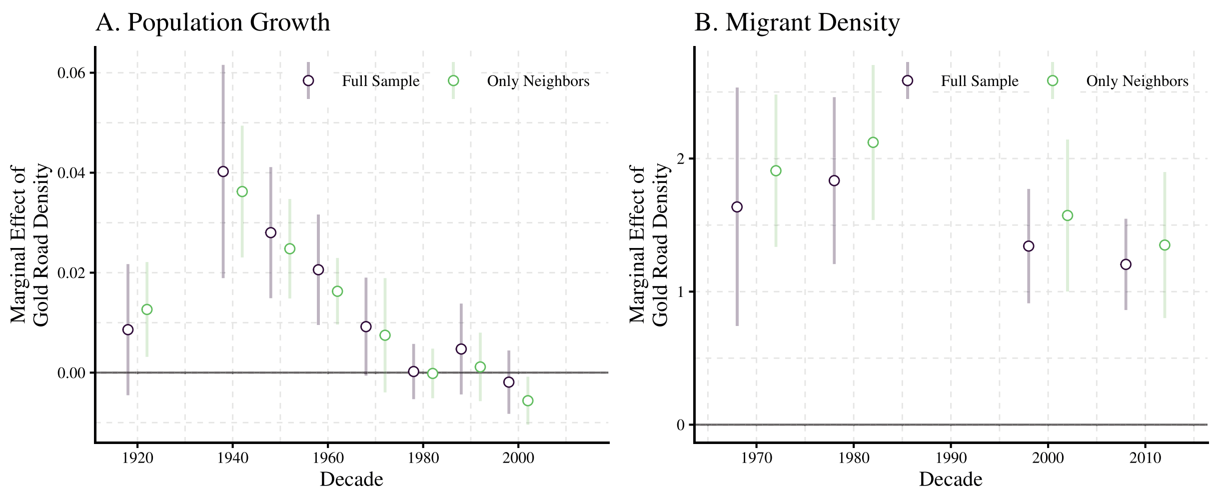

Fig. 6. Population growth and migration.
Notes: Panel A illustrates the effect of gold roads on average yearly population growth per decade, conditional on the initial population density. In 1920, it represents the average yearly population growth between 1920 and 1940. Panel B shows the effect of gold roads on the density of migrants from other municipalities. The *Full Sample* consists of 1920 MCAs, excluding those in states heavily influenced by the Amazon River basin and MCAs with settlements predating 1694 (except in years before 1694 in the shaded area), resulting in 856 observations. The *Only Neighbors* sample includes only MCAs traversed by gold roads and their immediate neighbors, amounting to 604 observations. All estimates incorporate state fixed effects and geography controls. In all cases, the Kleibergen–Paap F is above 76. Point estimates are represented by circles, with 95% spatially corrected confidence intervals (Conley, 1999) shown as lines. Exact values are detailed in Tables E.1 and E.2.

from regressions using 1920 MCAs, excluding those influenced by the Amazon River and those with settlements predating 1694. All estimates control for geography and state fixed effects, with standard errors spatially corrected following Conley (1999).

The coefficients presented in Fig. 6A indicate that higher gold road density increased population growth during the decades between 1940 and 1970. The effect between 1920 and 1940 was also positive but much smaller, likely due to the surge in industrialization in the 1930s. In the 1940s, a 10-percentage-point increase in gold road density implied approximately a 0.4-percentage-point increase in population growth. However, this effect diminished over the century, becoming negligible after 1980. The faster population growth between 1940 and 1980 contributed to a higher migrant density, as shown in Fig. 6B (for migrants, data is only available from 1970). Gold road density is associated with more residents born in other states and municipalities per square kilometer, suggesting that the road towns did, in fact, attract migrants.

More details are provided in Appendix E. Concerning gold road towns, Table E.3 shows that gold roads are associated with higher urbanization rates in all years for which we have data (1940 to 2010). These results show that gold road density is linked to modern settlements, and migration did not disrupt this pattern. However, additional results in Appendix G reveal a stark contrast between gold roads and mule roads. The relationship between mule roads and modern labor composition in 1920 is weaker than that of gold roads. The only coefficient statistically significant at the 10% level is associated with the share of workers in manufacturing. Furthermore, population growth is not significantly affected by mule roads until 1950, and the effect on migrant density from other states is either statistically insignificant or negative. These findings reinforce the hypothesis that mule roads were of secondary importance, as we exclude MCAs containing municipalities from 1872 from the sample.

In general, the results presented in this section demonstrate how historical roads, particularly gold roads, laid the foundation for agglomeration during demographic and structural transitions. Settlements along these roads were characterized by prototypical cities and attracted workers from other regions. In the mid-twentieth century, rapid urbanization was nationally attributed equally to natural population growth and internal migration (Wagner and Ward, 1980). In contrast, our findings reveal that areas with historical roads grew faster and had a larger stock of migrants.

7. Path dependence

At this stage, we have documented a distinctive dynamic pattern for the Brazilian economy. Historical roads constituted an initial shock to Brazil's spatial system, one that dissipated by the late nineteenth century. These roads, however, left a lasting imprint through the formation of road towns, which later positioned these regions to benefit from demographic and structural transitions. This pattern naturally raises a central question about the nature of Brazil's spatial system: Do long-run spatial imprints reflect mechanically slow mean reversion in a unique steady state, or can history matter through equilibrium selection and path dependence in a system that admits multiple steady states? In this section, we interpret the observed dynamics through the lens of spatial economics to shed light on this question and quantify the strength of agglomeration spillovers in this spatial system to answer whether they are consistent with a unique steady state or multiple steady states and path dependence.

This section uses the framework in Allen and Donaldson (2020, 2022) to answer this question. Our goal is not to replicate the full historical dynamics presented previously in a quantitative simulation, but to (i) estimate agglomeration spillovers from a wage–density system and (ii) use the implied spillovers to characterize whether the spatial system is likely to return to an original steady state where gold roads are not associated with more population, or whether multiple steady states and path dependence are plausible. Thus, we do not seek to extend the model to include multiple sectors. Instead, we use a single-sector model in which structural transformation raises aggregate productivity economy-wide, while former road towns may experience a temporary relative amenity advantage because their inherited urban functions make them especially attractive destinations for workers and migrants. If agglomeration forces are sufficiently strong, that temporary relative shock can push the economy toward a different long-run spatial configuration. In this sense, the model serves as a diagnostic: it asks whether our estimated agglomeration spillovers are strong enough to make equilibrium selection possible, not whether the gold-era dynamics themselves can be replicated quantitatively.

Model. The model contains key elements of two traditional strands of literature in spatial economics, including spatial linkages through trade and migration, agglomeration forces, realistic geographies, and forward-looking agents.[25] Specifically, each region i produces a unique variety of final good Y_{it} in period t using labor L_{it} and given the labor productivity A_{it}. Trade between i and j faces symmetric iceberg costs τ_{ijt}. Individuals draw utility from a Dixit-Stiglitz love-for-variety aggregator with elasticity of substitution between varieties denoted by σ. Moreover, they enjoy local amenities u_{it} and benefit from higher utility from future generations (forward looking), discounted by the parameter δ. Individuals have idiosyncratic preferences over locations, which are extreme value distributed with shape parameter θ.

Both local productivity and local amenities are assumed to be affected by current and historical population. Specifically, $A_{it} = \bar{A}_{it} L_{it}^{\alpha_1} L_{it-1}^{\alpha_2}$, where α_1 and α_2 represent the strength of contemporaneous and historical productivity spillovers, respectively, and $u_{it} = \bar{u}_{it} L_{it}^{\beta_1} L_{it-1}^{\beta_2}$, where β_1 and β_2 denote the strength of contemporaneous and historical amenity spillovers, respectively. Individuals' and firms' optimization yields an extended version of the labor supply and demand system in the Rosen–Roback–Glaeser spatial equilibrium tradition, where lagged population acts as a shifter in both equations:

$$\log w_{it} = \left(\alpha_1 \frac{\sigma-1}{\sigma} - \frac{1}{\sigma}\right) \log L_{it} + \left(\alpha_2 \frac{\sigma-1}{\sigma}\right) \log L_{it-1} + \frac{1}{\sigma} \log Q_{it}^{1-\sigma} + \frac{\sigma-1}{\sigma} \log \bar{A}_{it}, \tag{2}$$

$$\log w_{it} = \left(\frac{1}{\theta} - \beta_1\right) \log L_{it} + (-\beta_2) \log L_{it-1} + \frac{1}{\theta} \log \Lambda_{it}^{\theta} + \frac{1}{1-\sigma} \log P_{it}^{1-\sigma} - \frac{\delta}{\theta} \log \Pi_{it}^{\theta} - \log \bar{u}_{it}, \tag{3}$$

where w_{it} is the nominal wage, $P_{it}^{1-\sigma}$ and $Q_{it}^{1-\sigma}$ are the inward and outward market access, respectively, and Λ_{it}^{θ} and Π_{it}^{θ} are the inward and outward migration market access, respectively. See Appendix F (or Allen and Donaldson, 2020) for details. Now, to facilitate the interpretation of the system's dynamical properties, we follow Allen and Donaldson (2022) and simplify the environment with a homogeneous good such that there is no trade ($\sigma \to \infty$) and endogenous fertility offsets the negative effect of outward migration market access. The resulting supply and demand system in the simplified environment is

$$\log w_{it} = \alpha_1 \log L_{it} + \alpha_2 \log L_{it-1} + \log \bar{A}_{it}, \tag{4}$$

$$\log w_{it} = \left(\frac{1}{\theta} - \beta_1\right) \log L_{it} + (-\beta_2) \log L_{it-1} + \frac{1}{\theta} \log \Lambda_{it}^{\theta} - \log \bar{u}_{it}. \tag{5}$$

Throughout, the simplified no-trade model serves to build analytical intuition and to derive closed-form conditions for each regime. The trade-augmented model in Eq. (2) and (3) — which incorporates the inward and outward market-access measures constructed in Appendix F — is our preferred quantitative specification. We perform the diagnostic with both models and reach similar conclusions: the estimated spillovers place the spatial system in a region where multiple steady states cannot be ruled out.

[25] One strand focuses on path-dependent geographies, agglomeration forces, and forward-looking behavior, but typically in low-dimensional settings (*e.g.* Krugman, 1991; Matsuyama, 1991; Rauch, 1993). The other involves high-dimensional models that incorporate more realistic geographies, though they often lack forward-looking agents and local agglomeration spillovers (*e.g.* Roback, 1982; Glaeser, 2008; Desmet et al., 2018).

Solving the system formed by Eq. (4) and (5) for all L_{it} yields the key law of motion:

$$L_{it}^{1-\theta(\alpha_1+\beta_1)} = (\bar{A}_{it}\bar{u}_{it})^{\theta} \times L_{it-1}^{\theta(\alpha_2+\beta_2)} \times \sum_j \mu_{jit}^{-\theta} L_{jt-1}, \tag{6}$$

where we use $\Lambda_{it}^{-\theta} = \sum_j \mu_{jit}^{-\theta} L_{jt-1}$ to emphasize that inward migration market access depends on lagged population in all regions. The geography of the system, composed of $\bar{A}_{it}$, $\bar{u}_{it}$, and migration costs μ_{ijt} for all i, j, and t, as well as the initial population distribution, is taken as exogenous.

In much of the historical persistence literature, "persistence" is a reduced-form statement: A past shock that leaves a detectable imprint even centuries later. In the model, such long-lived effects can arise through two distinct mechanisms. First, the system may have a unique steady state but converge slowly (high within-basin persistence). Second, the system may admit multiple steady states so that history affects equilibrium selection (path dependence). Following Allen and Donaldson (2022), we therefore distinguish three concepts:

Equilibrium-path uniqueness and stability: Given geography and an initial population distribution, Eq. (6) pins down a unique equilibrium path $\{L_{it}^*\}_{t=1}^{\infty}$ for all i as long as $\theta(\alpha_1+\beta_1) \neq 1$. The equilibrium path is locally stable if $\theta(\alpha_1+\beta_1) < 1$. Importantly, this condition depends only on contemporaneous agglomeration forces.

Path dependence (multiple steady states): *Path dependence requires multiple steady states*. Under constant geography, steady states satisfy

$$(L_i^{ss})^{1-\theta(\alpha_1+\alpha_2+\beta_1+\beta_2)} = \sum_j \left(\frac{\bar{A}_i\bar{u}_i}{\mu_{ji}}\right)^{\theta} L_j^{ss}.$$

This mapping has a unique fixed point under a contraction-type condition, where a sufficient condition in the simplified environment is $\left|[1-\theta(\alpha_1+\alpha_2+\beta_1+\beta_2)]^{-1}\right| < 1$. When this condition fails, multiple steady states may exist, and temporary shocks can have permanent consequences through equilibrium selection.

Persistence (speed of convergence within a basin): [26]Conditional on stability, "persistence" refers to the speed at which the system converges back to a steady state after a temporary shock *absent a basin shift*. The intuition is clearest in partial equilibrium, where inward migration market access is treated as exogenous (appropriate for sufficiently small regions whose shocks have limited effects on aggregate migration incentives). In this case, a log-linearized version of Eq. (6) has an AR(1)-type propagation with persistence parameter

$$\rho^{PE} = \frac{\theta(\alpha_2+\beta_2)}{1-\theta(\alpha_1+\beta_1)}.$$

Assuming a stable equilibrium path ($\theta(\alpha_1+\beta_1) < 1$), *partial-equilibrium convergence* holds when $\rho^{PE} < 1$, equivalently $\theta(\alpha_1+\alpha_2+\beta_1+\beta_2) < 1$. Allen and Donaldson (2022) also derive the general-equilibrium persistence parameter

$$\rho^{GE} = \frac{1+\theta(\alpha_2+\beta_2)}{1-\theta(\alpha_1+\beta_1)},$$

and show that *uniform convergence* (global convergence to the same steady state from any initial condition) requires sufficiently weak net agglomeration forces (in the simplified environment, $\theta(\alpha_1+\alpha_2+\beta_1+\beta_2) < 0$). Thus, if $\theta(\alpha_1+\alpha_2+\beta_1+\beta_2) > 1$ then there is *divergence* between regions, in the sense that the effect of a historical shock can be felt many years later even without multiple steady states.

[26] A *basin of attraction* refers to the set of initial population distributions from which the system converges to a given steady state, following Allen and Donaldson (2022, p. 20).

Interpretation of the historical pattern. Through the lens of this model, the observed historical pattern can be interpreted as a spatial system that experienced two shocks. First, the discovery of gold increased the productivity of mining sites and those along the gold roads, thereby constituting a labor demand shock. With gold depletion, productivity advantages dissipated, and the system returned to the initial steady state, in which gold road locations do not exhibit disproportionately higher population density. This return to the initial steady state is consistent with two possible parameter regions: one in which the system has a unique and stable steady state, and one in which multiple steady states are possible but the shock was not large enough to shift the economy to a different basin of attraction. It is important to note that our subsequent quantitative analysis cannot distinguish between these two cases unless we assume that the parameters estimated from the twentieth-century data are the same as those governing the dynamics during the gold rush.

The second shock results from a combination of aggregate-level demographic and structural changes and local advantages conferred by gold roads. The mechanism of structural changes, which can be captured by a permanent, economy-wide increase in aggregate productivity, induces large-scale rural–urban migration. Road towns disproportionately attracted these migrants due to their early urban function, which we interpret as an amenity advantage. Since the gold rush era, road towns possessed relatively higher urban amenities compared to the dispersed agricultural settlements that surrounded them—commerce, accommodation, civic services, and a non-agricultural population mix. But, as mentioned in Section 6, these amenities conferred no meaningful advantage in a sparse agrarian economy: When location choice was governed primarily by proximity to cultivable land, urban services were largely irrelevant to where workers settled. The amenity advantage was real but dormant. Structural transformation changed the relevant margin of location choice. As structural transformation freed workers from agriculture and generated large-scale rural–urban migration, workers began choosing among urban destinations based on urban amenities and services rather than agricultural land quality (Wagner and Ward, 1980). Road towns' pre-existing $\bar{u}_i$ advantage became operative during this transition—it is not that their amenities improved, but the transition made amenities the binding factor for location choice.

Since we do not extend the model to include multiple types of amenities and sectors, we interpret the second shock as a temporary elevation of intrinsic amenities $\bar{u}_i$ for road towns during the urbanization transition—an approximation of the deeper regime shift, from land-governed to amenity-governed location choice, that a single-sector model cannot fully represent. This interpretation is consistent with the re-emergence of gold road effects observed after 1920.

This advantage is temporary because urban amenities are replicable: Commerce, accommodation, civic infrastructure, and service industries can be built anywhere, given sufficient population and economic activity. Road towns had a first-mover advantage—they had already developed this infrastructure when structural transformation began, whereas other localities had not. As development spread and other localities received investment and population growth, they too developed comparable urban functions, and the relative $\bar{u}_i$ gap closed. This contrasts with a geographic fundamentals story (a natural harbor, a river crossing, a mineral deposit), where the advantage is non-replicable and permanent. Road towns hold no permanent monopoly on commerce or services; their advantage was temporal, not structural. This is reflected in the decrease in differential population growth over time, as shown in Fig. 6A.

Whether the temporary amenity advantage was sufficient to cross a basin boundary depends on the strength of agglomeration spillovers. If spillovers are sufficiently strong, the shock can shift the economy to a different long-run spatial configuration where gold roads are associated with higher population density. If spillovers are weak, the system should return to the original steady state, in which gold roads do not confer a long-run population advantage. The next step is to estimate these spillovers and use them to characterize the spatial system's equilibrium properties during the twentieth century.

Estimation. Next, our goal is to estimate the net agglomeration spillovers and use them to locate Brazil's spatial system in the regime map implied by the model and definitions above. We estimate the following supply and demand system:

$$\log w_{it} = \zeta_1 \log L_{it} + \zeta_2 \log L_{it-1} + \zeta_3 \log Q_{it}^{1-\sigma} + \epsilon_{it}, \tag{7}$$

$$\log w_{it} = \nu_1 \log L_{it} + \nu_2 \log L_{it-1} + \nu_3 \log \Lambda_{it}^{\theta} + X_{it}'\nu_4 + \omega_{it}. \tag{8}$$

In this system, wages and population density are jointly determined. To identify the productivity spillovers in labor demand Eq. (7), we require an instrument that shifts labor supply but is excluded from labor demand. We use the LCPs of historical roads as this excluded instrument for contemporaneous density.[27]

The identifying assumption is that, as discussed above, historical roads shifted the spatial distribution of population by creating urban-type settlements that, during the process of structural transformation, eventually attracted workers, operating as a shock to the amenities term that shifts labor supply. Thus, the instrument provides exogenous variation in contemporaneous density that traces out the labor demand schedule, allowing us to identify the elasticity ζ_1. Conditional on lagged density, geography, individual controls, and (in the trade specification) outward market access, historical-road LCP exposure does not affect wages except through its effect on contemporaneous density.

Several features of our specification support this exclusion restriction. First, we control flexibly for worker characteristics, including industry and education, so identification comes from within-industry and within-skill wage differences associated with exogenous variation in density. This restriction does not mean that historical roads have no effect on intermediate outcomes. Indeed, Table 3 documents that gold-road exposure predicts sectoral shifts away from agriculture toward manufacturing and services in 1920. These compositional effects are absorbed by our controls: We identify productivity spillovers from the residual wage premium that density confers on observationally equivalent workers.

Second, by controlling for lagged density (L_{it-1}), we absorb the persistent productivity-relevant features of historical urbanization. We emphasize that lagged population density enters the supply and demand equations as a *state variable* in the underlying dynamic system (see Eq. (6)), not as an instrument for contemporaneous density. This distinguishes our approach from the urban-economics tradition following Ciccone and Hall (1996), in which historical population is used as an excluded instrument; here, our excluded instrument is the least-cost-path exposure, and lagged density is a conditioning variable that absorbs historical persistence. A concern might arise that historical roads, by generating urban settlements, created durable agglomeration economies that affect productivity independently of contemporaneous density. Lagged density captures the historical component of this variation. Additionally, measuring lagged density in 1920 provides at least a 60-year separation between the historical and contemporary periods, so identification comes from the instrument's effect on density changes since 1920. Third, the economic rationale for roads devised for colonial mineral extraction is orthogonal to the determinants of modern

[27] One alternative approach would be to estimate the reduced-form log version of Eq. (6), in which the coefficients on $\log \Lambda_{it}^{-\theta}$ and $\log L_{it-1}$ map directly to $[1-\theta(\alpha_1+\beta_1)]^{-1}$ and $\theta(\alpha_2+\beta_2)[1-\theta(\alpha_1+\beta_1)]^{-1}$, respectively. While feasible, this approach imposes stronger exclusion restrictions than the structural system because the reduced form collapses the supply and demand equations into a single composite error term. Moreover, the resulting reduced-form estimates imply agglomeration parameters that are implausibly large relative to the empirical literature (for example, $\alpha_1 \approx 0.38$ and $\alpha_2 \approx -0.14$ under benchmark values for β_1 and β_2), placing the system near the boundary of partial-equilibrium non-convergence. For these reasons, and because identification is cleaner when focusing on the supply equation alone, we base our inference on the structural system rather than the reduced form. Full estimation results are available upon request.

manufacturing and service-sector productivity, making direct effects on contemporaneous production technology less plausible.

The generic system in Eq. (7) and (8) nests both systems above. In the simplified environment Eq. (4) and (5), we set $\zeta_3 = 0$, implying $\alpha_1 = \zeta_1$ and $\alpha_2 = \zeta_2$. In the environment with trade Eq. (2) and (3), we either restrict $\zeta_3 = 1/\sigma$ or estimate ζ_3; in both cases we recover

$$\alpha_1 = (\zeta_1 + 1/\sigma)\frac{\sigma}{\sigma - 1}, \qquad \alpha_2 = \zeta_2 \frac{\sigma}{\sigma - 1}.$$

We do not estimate β_1 and β_2 directly. Instead, we interpret equilibrium properties under feasible values of $\beta_{ss} \equiv \beta_1 + \beta_2$. As discussed by Allen and Donaldson (2022), β_1 is expected to be negative if there is a fixed supply of local goods (e.g., land or housing congestion), while β_2 is expected to be positive if early investments in local factors are durable. Using evidence on housing expenditure shares and assuming housing durability over a generation, we adopt the benchmark $\beta_1 = -\beta_2 = -0.15$ for Brazil (Chauvin et al., 2017). In their U.S. application, Allen and Donaldson (2020) estimate $\beta_1 = -0.26$ and $\beta_2 = 0.31$, implying β_{ss} close to zero. We follow Allen and Donaldson (2020) setting $\theta = 4$, $\sigma = 9$, and $\delta = 0.03$.

Consistent with the literature (*e.g.,* Bleakley and Lin, 2012; Chauvin et al., 2017), we use individual-level census data on workers aged 25 to 65 to estimate agglomeration effects. We have all the information needed for the 1980, 2000, and 2010 censuses. Hourly wages are set as the dependent variable, while contemporaneous and historical population density are computed at the 1920 MCA level, along with the usual geography variables. Lagged population density is measured 60 years prior to the census year. Outward market access $Q_{it}^{1-\sigma}$ was computed following Allen and Donaldson (2020). We construct a 5×5 km conductivity grid for Brazil combining information on roads, railways, and waterways, and compute LCPs using the Fast Marching Method (Allen and Arkolakis, 2014). The iceberg costs are calibrated by estimating the elasticity of trade flows to freight costs using state-level data from Morten and Oliveira (2024). Details are presented in Appendix F. All regressions include individual-level controls such as age, age squared, and binary variables indicating the worker's race, sex, marital status, industry, and attained education. Table 4 shows the results pooling all censuses and adding state and year fixed effects.[28]

Before turning to the estimates, it is useful to clarify the structure of Table 4. Columns (1)–(4) report the simplified model, which shuts down spatial linkages by imposing $\zeta_3 = 0$; accordingly, outward market access does not enter those specifications. Column (1) includes only contemporaneous population density, Column (2) adds lagged population density, and Columns (3) and (4) add geography controls using the full sample and neighbors-only sample, respectively. Columns (5) and (6) report the trade model, where outward market access enters through $Q_{it}^{1-\sigma}$. Column (5) estimates ζ_3 freely, while Column (6) imposes the structural restriction $\zeta_3 = 1/\sigma$ with $\sigma = 9$.

The elasticity of wages with respect to contemporaneous population density is stable across specifications, ranging from 0.098 to 0.124. In the simplified environment and our preferred specification, we estimate contemporaneous and historical productivity spillovers of 0.103 (0.028) and −0.03 (0.021), respectively, implying total productivity spillovers $\alpha_{ss} = \alpha_1 + \alpha_2 = 0.073$ (0.035). This magnitude is close to the typical range in the literature (0.03–0.08) summarized by Rosenthal and Strange (2004) and Combes and Gobillon (2015), and to the estimate in Bleakley and Lin (2012). Our estimates are somewhat larger than those reported in other papers on Brazil (e.g., 0.026 in Chauvin et al., 2017 and 0.01 in Ehrl and Monasterio, 2021). A natural interpretation is that our instrument captures variation linked to road towns that were comparatively more likely to generate larger productivity spillovers in a country where settlement was historically dominated by dispersed agricultural estates. Table 5 summarizes the parameters estimated in this paper, together with those assumed from the literature, which we use in the following numerical study of the system's properties.

In Fig. 7, we plot equilibrium regions of the simplified environment for different combinations of $\alpha_{ss} \equiv \alpha_1 + \alpha_2$ and $\beta_{ss} \equiv \beta_1 + \beta_2$ (with $\theta = 4$). Circles denote our point estimate of α_{ss} from Column (3) of Table 4 evaluated at $\beta_{ss} = 0$, and the dashed rectangle reports a conservative confidence region allowing α_{ss} to vary within ± 1.96 standard errors of the point estimates and $\beta_{ss} \in [-0.05, 0.05]$.

We highlight three results regarding the implied properties of Brazil's spatial system. First, for our benchmark contemporaneous estimates $\alpha_1 \approx 0.1$ and $\beta_1 = -0.15$, we have $\theta(\alpha_1 + \beta_1) < 1$, implying a unique and stable equilibrium path in the simplified dynamics. Second, our estimate $\alpha_{ss} \approx 0.07$ implies that for a broad range of feasible β_{ss} values we lie in the region $0 < \theta(\alpha_{ss} + \beta_{ss}) < 1$. This guarantees convergence in partial equilibrium while ruling out uniform convergence. In economic terms, net agglomeration forces are strong enough that the model does not necessarily converge to the same steady state from all initial conditions (so multiple steady states remain possible), but not strong enough to generate non-mean-reverting dynamics (divergence).

Third, to quantify within-basin persistence, we compute ρ^{PE} using $\alpha_2 = -0.03$ and $\beta_2 = 0.15$, we obtain $\rho^{PE} \approx 0.4$, implying relatively rapid decay: Absent a basin shift, less than 6% of an initial shock would remain after two centuries. Hence, the long-run historical persistence patterns we document are more consistent with equilibrium selection and the interaction between early settlement seeds and later aggregate changes than with mechanically slow mean reversion alone. These implications are robust across a wide range of feasible β_{ss} values. Only when β_{ss} is close to −0.05 can the system fall into the uniform convergence region, which would imply a unique steady state with global convergence.

Turning to the environment with trade, in our preferred specification in Column (6) of Table 4, the implied contemporaneous and historical agglomeration forces are 0.242 (0.032) and −0.05 (0.023), respectively. Our estimates are somewhat larger but close to those of Allen and Donaldson (2020), who find $\alpha_1 = 0.19$ and $\alpha_2 = -0.041$. As in the simplified environment, our estimates satisfy $\theta(\alpha_1 + \beta_1) < 1$, implying a unique and stable equilibrium path. Finally, the estimated coefficient on outward market access in Column (5) implies $\sigma \approx 10.2$, close to the value we use in our calibration and common in the quantitative spatial literature.

To assess whether the trade-augmented system admits a unique steady state or multiple steady states, we apply our candidate steady states to the condition derived in Allen and Donaldson (2020), which shows that multiple steady states and path dependence cannot be ruled out. Appendix F reports the numerical implementation. Overall, the simplified and trade-augmented models deliver consistent qualitative implications: stability of the equilibrium path together with the possibility that history affects long-run outcomes through equilibrium selection.

The results in this section suggest that the Brazilian spatial system is in a regime where multiple steady states are possible, and thus there is scope for temporary shocks to have long-run effects through equilibrium selection. However, whether the shock observed in the twentieth century was large enough to shift the economy to a different basin of attraction cannot be determined with certainty. This exercise rules out a scenario in which return to the initial steady state would be inevitable, while leaving open whether the twentieth-century shock was large enough to cross a basin boundary or whether the system will

[28] As a robustness check, we also estimated the agglomeration parameters separately for 1980, 2000, and 2010, rather than pooling years. The resulting estimates of α_1 — 0.140 (0.022), 0.086 (0.032), and 0.060 (0.040), respectively — are all statistically indistinguishable from the pooled value of 0.105 (0.030), with substantially overlapping confidence intervals. The modest variation across decades is driven primarily by differences in precision, especially in later years where the identifying variation is weaker. Importantly, all decade-specific estimates lie well within the region of the parameter space consistent with the possibility of multiple steady states. Full regression tables are available upon request.

Table 4
Contemporaneous and historical productivity spillovers.

	Simple model				Model with trade	
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable: Hourly Wage						
Population Density	0.100***	0.124***	0.098***	0.103***	0.105***	0.105***
	(0.012)	(0.024)	(0.034)	(0.028)	(0.028)	(0.029)
Lagged Pop. Density		−0.049*	−0.018	−0.030	−0.042**	−0.044**
		(0.026)	(0.031)	(0.021)	(0.019)	(0.021)
Outward Market Access					0.098**	
					(0.044)	
Observations	11,039,347	11,039,347	11,039,347	8,957,715	8,957,715	8,957,715
Kleibergen–Paap F	44.630	23.335	20.136	23.687	21.229	23.687
Individual Controls:	✓	✓	✓	✓	✓	✓
Geography Controls:			✓	✓	✓	✓
Only Neighbors:				✓	✓	✓
$\zeta_3 = 1/\sigma$						✓

Notes: Spatially corrected standard errors are shown in parentheses. All columns report 2SLS estimates, using the LCP density of gold roads as an IV for population density at the 1920 MCA level. The *Full Sample* consists of individual data for years 1980, 2000, and 2010, excluding those in states predominantly influenced by the Amazon River basin and those in MCAs with settlements predating 1694. The *Only Neighbors* sample includes only individuals in MCAs traversed by historical roads and their immediate neighbors. Each specification includes state and year fixed effects and individual characteristics such as age (and age squared) and binary variables indicating race, sex, industry, marital status, and attained education. Geography controls include the log-transformed median values of precipitation, temperature, elevation, terrain ruggedness, area, distance to rivers, and distance to the coast, as well as a second-order latitude–longitude polynomial. *Hourly Wage* refers to the log-transformed ratio of wages to hours worked. *Pop. Density* is defined as the log-transformed value of population divided by the MCA's area. *Lagged Pop. Density* is the log-transformed value of population density in 1920. *p<0.1,**p<0.05,*** p<0.01.

Table 5
Summary of parameters.

Parameter	Description	Value	Notes
Panel A - Simple Model			
$\hat{\alpha}_1$	Contemporaneous productivity spillover	0.103 (0.028)	Estimated.
$\hat{\alpha}_2$	Historical productivity spillover	−0.030 (0.021)	Estimated.
$\widehat{\alpha_1 + \alpha_2}$	Total productivity spillover	0.073 (0.035)	Implied from estimates. Within literature review range 0.03–0.08 (Rosenthal and Strange, 2004; Combes and Gobillon, 2015) and 0.09 (Bleakley and Lin, 2012). For Brazil, 0.01 (Ehrl and Monasterio, 2021) and 0.026 (Chauvin et al., 2017).
Panel B - Trade Model			
$\hat{\alpha}_1$	Contemporaneous productivity spillover	0.242 (0.032)	Estimated. Comparable to 0.19 in Allen and Donaldson (2020).
$\hat{\alpha}_2$	Historical productivity spillover	−0.05 (0.023)	Estimated. Comparable to −0.041 in Allen and Donaldson (2020).
$\widehat{\alpha_1 + \alpha_2}$	Total productivity spillover	0.192	Implied from estimates. Comparable to 0.149 in Allen and Donaldson (2020).
δ	Intertemporal discount factor	0.03	To match the average annualized return on wealth in the US equal to 0.0603 (Allen and Donaldson, 2020).
Panel C - Both Models			
β_1	Contemporaneous amenity/congestion spillover	−0.15	To match an unmodeled housing expenditure share equal to 15% (Chauvin et al., 2017) as in Allen and Donaldson (2022).
β_2	Historical amenity/durability spillover	0.15	To match an unmodeled housing fully durable over a generation as in Allen and Donaldson (2022).
θ	Taste dispersion/migration sensitivity	4	To match the location choice elasticity estimated by Monte et al. (2018), adjusted to a dynamic framework as in Allen and Donaldson (2020).
σ	Elasticity of substitution	9	To match the trade elasticity estimated by Donaldson and Hornbeck (2016).

eventually return to its original steady state. Furthermore, the parameters governing the system's dynamics during the colonial period may differ from those estimated in the twentieth century, so interpreting the dynamics prior to 1920 using the findings of this section requires caution and is beyond the scope of this analysis.

8. Conclusions

In this paper, we examined the long-run impact of historical roads on Brazil's population distribution. Specifically, we analyzed the construction of pathways connecting the Brazilian coast to the gold-mining

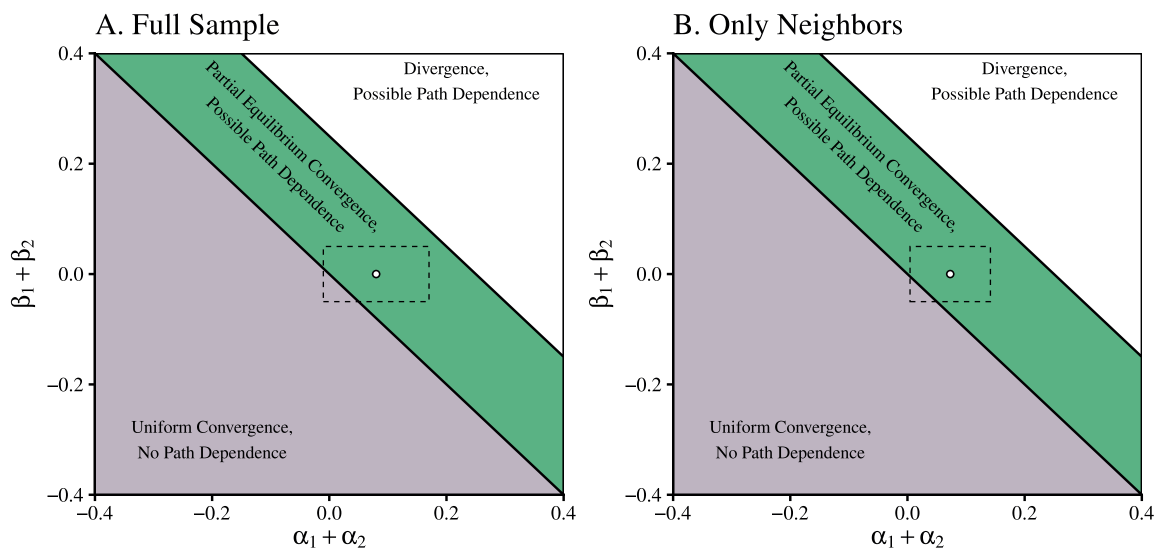

Fig. 7. Equilibrium properties and spillover estimates.
Notes: Both panels illustrate the equilibrium properties of the model for different combinations of productivity ($\alpha_{ss} \equiv \alpha_1+\alpha_2$) and amenity ($\beta_{ss} \equiv \beta_1+\beta_2$) spillovers. Circles represent point estimates of α_{ss} in Column (3) of Table 4 (Panels A and B) assuming $\beta_{ss}=0$, conditional on the initial population density. The dashed line rectangle represents a confidence region where α_{ss} ranges within ± 1.96 standard errors and $\beta_{ss} \in [-0.05, 0.05]$.

regions of the interior following the discovery of gold around 1694. We found that the density of these historical routes is a strong predictor of modern agglomeration, as measured by population density, urban population density, and nightlight intensity.

Our analysis uncovered a distinctive dynamic pattern. Initially, historical roads facilitated population growth, but by the late nineteenth century, their influence on agglomeration had largely disappeared. However, following Brazil's demographic and industrial transitions after the 1930s, locations with higher historical road density experienced faster population growth and attracted a larger share of migrants. This suggests that the road towns established along these routes played a central role in shaping future urbanization and economic activity.

Throughout the twentieth century, the impact of historical roads strengthened, indicating that these early transportation networks became increasingly important for the spatial distribution of population. To interpret this pattern, we estimated net agglomeration effects in a dynamic spatial framework and found that the implied forces are consistent with an environment with multiple steady states in which temporary spatial shocks can have permanent consequences through equilibrium selection. In Brazil's case, history matters not because the gold discovery shock was sustained indefinitely, but because the initial conditions it created may have interacted with later structural transformation to help select a long-run spatial equilibrium in which places along historical corridors retain a larger share of population and economic activity.

CRediT authorship contribution statement

Diogo Baerlocher: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Guilherme Lambais:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Eustáquio Reis:** Writing – review & editing, Data curation, Conceptualization. **Diego Firmino Costa da Silva:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Henrique Veras:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jue.2026.103863.

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